test area

Inhalt

- test area
  - load
  - pls
    - bars
    - histo
    - box
    - pie
    - join
    - sankey
  - tbl
    - print_summary()
    - describe_df()
    - descr_db()
  - show num
  - venn
  - hlp

load

> this is loading

```
🐍 3.12.8 | 📦 matplotlib_venn: 0.11.10 | 📦 dataframe_image: 0.2.7 | 📦
plotly: 6.2.0 | 📦 kaleido: 1.0.0 | 📦 seaborn: 0.13.2 | 📦 pandas: 2.3.3 |
📦 numpy: 1.26.4 | 📦 duckdb: 1.3.2 | 📦 pandas-plots: 1.4.0
True
```

```
    n = 3_241_401
    └ [2020-2023.07]:                    n = 2_633_644 (100.0%)
    └ [not z_is_dco]:                    n = 2_547_636  (96.7%)
    └ [keine M1]:                        n = 2_305_215  (87.5%)
    └ [keine Verstorbenen < 180 Tage]:   n = 2_132_064  (81.0%)
    └ [lympho- und mesoendokr. Tumore]:    n = 27_653   (1.0%)
```

▶ filter-sql

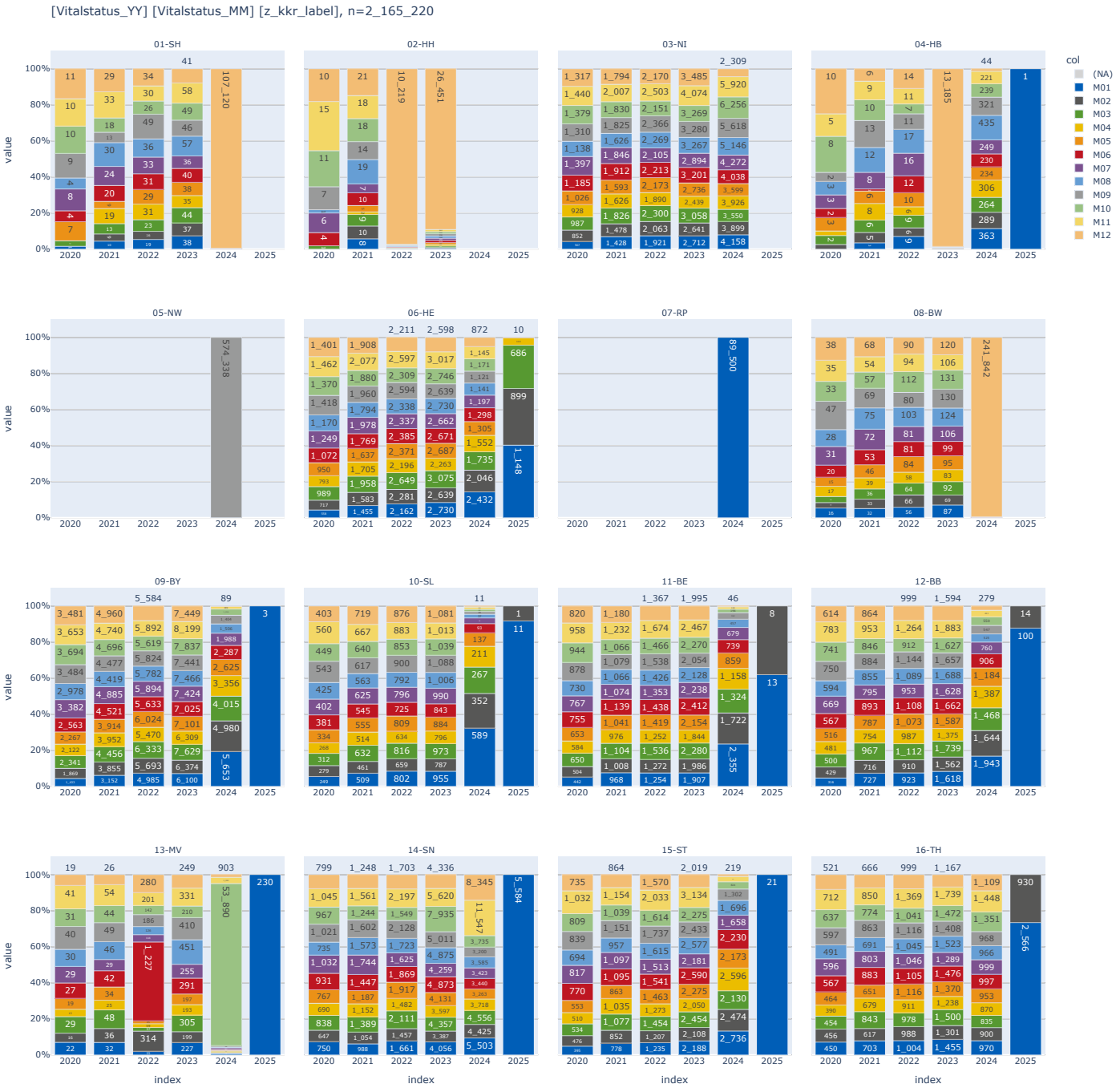
```
Diagnosedatum between '2020-01-01' and '2023-06-30'
and not z_is_dco
```

```
and ifnull(z_m_pc_1,'') <> '1'
and ifnull(z_period_diag_death_day,181) >= 180
and z_icd10 in ('C91.0', 'C92.0', 'C83.3', 'C82.4')
```

```
    count: distinct z_pat_id
    ---
    n = 2_735_152                           (100.0%)
    └ [DJ 2020-2024]:        n = 2_729_035  (99.8%)
    └ [Verstorben = 'N']:    n = 1_975_774  (72.2%)
    └ [Vitalstatus >= 2020]: n = 1_975_074  (72.2%)
```

▶ filter-sql

```
z_dy between 2020 and 2024
and Verstorben = 'N'
and year(Datum_Vitalstatus) between 2020 and 2025
```




```
count: distinct z_pat_id
---
n = 2_735_152
└ [2020–2023.07]:                    n = 2_415_210 (100.0%)
└ [not z_is_dco]:                    n = 2_334_551  (96.7%)
└ [keine M1]:                        n = 2_113_894  (87.5%)
└ [keine Verstorbenen < 180 Tage]:   n = 1_956_584  (81.0%)
└ [lympho- und mesoendokr. Tumore]:    n = 27_520   (1.1%)
```

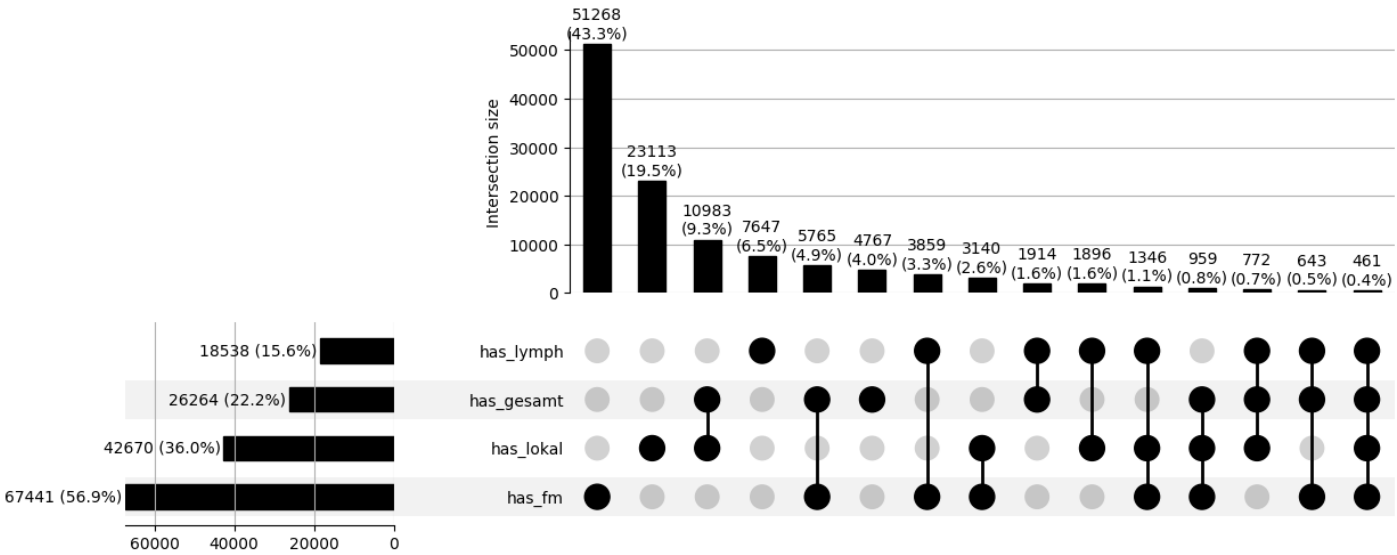
▶ filter-sql

```
Diagnosedatum between '2020-01-01' and '2023-06-30'
and not z_is_dco
and ifnull(z_m_pc_1,'') <> '1'
and ifnull(z_period_diag_death_day,181) >= 180
and z_icd10 in ('C91.0', 'C92.0', 'C83.3', 'C82.4')
```

```
🗄️ db   2_265_000, 4
    ("has_gesamt, has_lokal, has_lymph, has_fm")
```

has_gesamt	has_lokal	has_lymph	has_fm
false	false	false	false
false	NULL	NULL	NULL
false	false	false	false

```
n = 2_265_000 | n(true) = 152_567 | n(nonempty) = 118_533
```



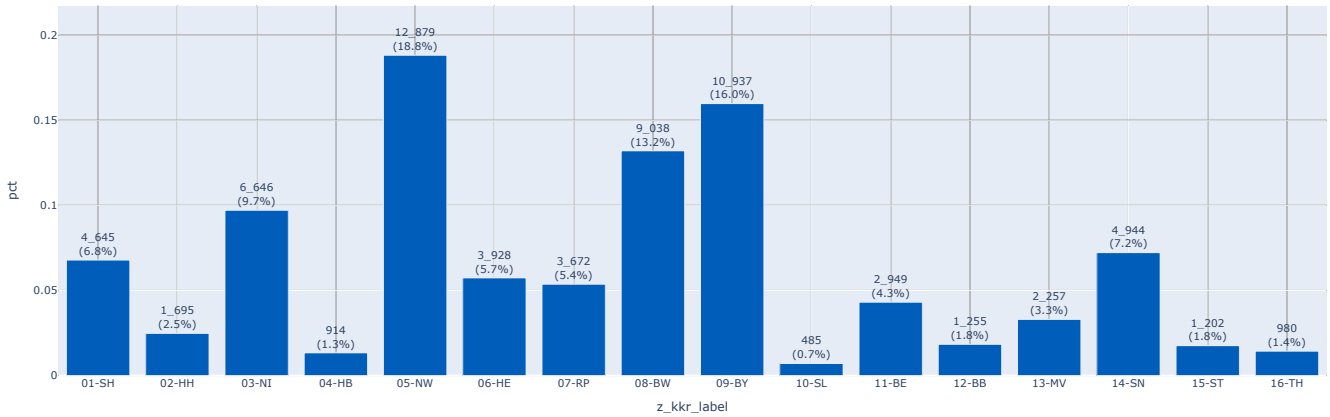
color payment	green	yellow	Total
(NA)	20 (0.0%)	508 (0.6%)	528 (0.6%)
cash	3_995 (4.7%)	17_012 (20.2%)	21_006 (24.9%)
credit card	9_774 (11.6%)	52_907 (62.8%)	62_681 (74.4%)
Total	13_788 (16.4%)	70_427 (83.6%)	84_215 (100.0%)

color payment	green	yellow	Total
(NA)	20 (0.0%)	508 (0.6%)	528 (0.6%)
cash	3_995 (4.7%)	17_012 (20.2%)	21_006 (24.9%)
credit card	9_774 (11.6%)	52_907 (62.8%)	62_681 (74.4%)
Total	13_788 (16.4%)	70_427 (83.6%)	84_215 (100.0%)

pls

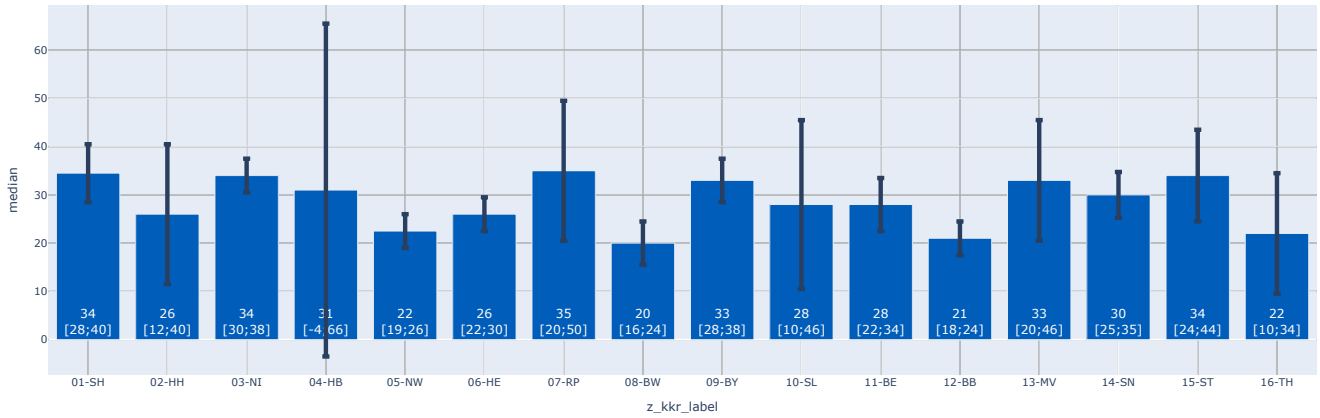
bars

[z_first_treatment_after_days] by [z_kkr_label], n=68_426

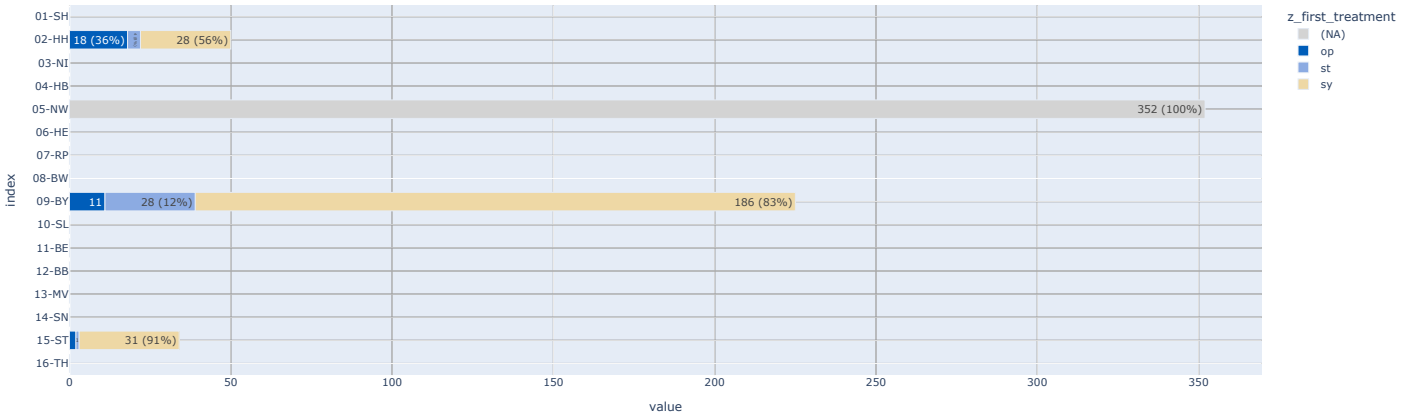


[z_first_treatment_after_days] by [z_kkr_label], NULL excluded, n=1_724)

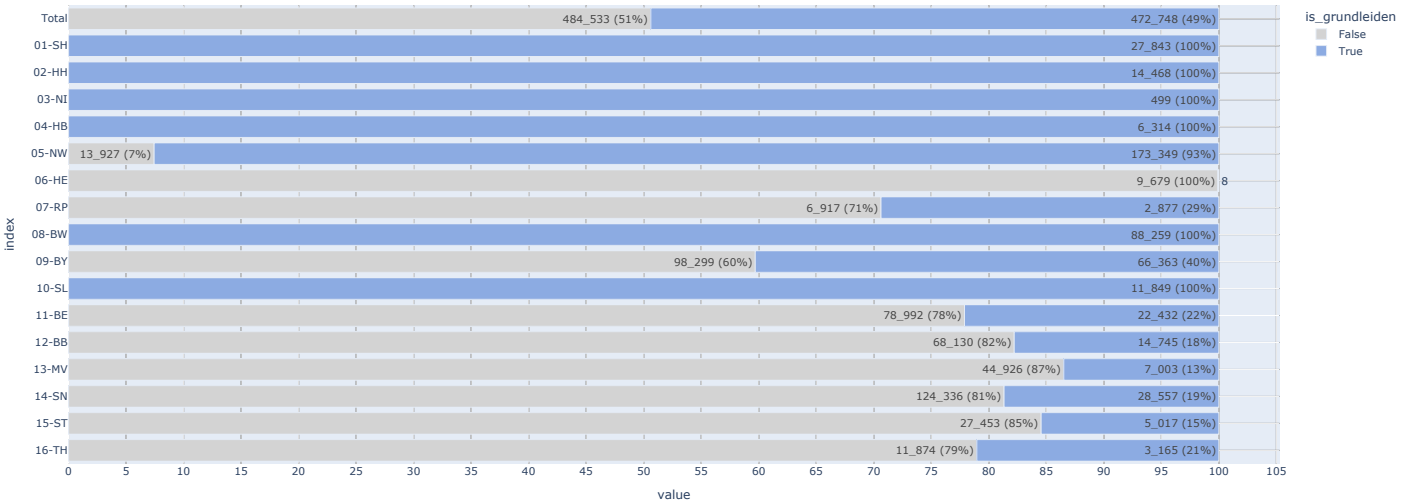
ci(95) on medians



pl, n=661



test, [kkr] by [is_grundleiden], n=957_281



	z_dy	tu	z_kkr_label	cnt
0	2020	06	13-MV	1
1	2020	30	13-MV	1
2	2020	###	13-MV	21
3	2020	18	11-BE	1
4	2020	18	13-MV	1

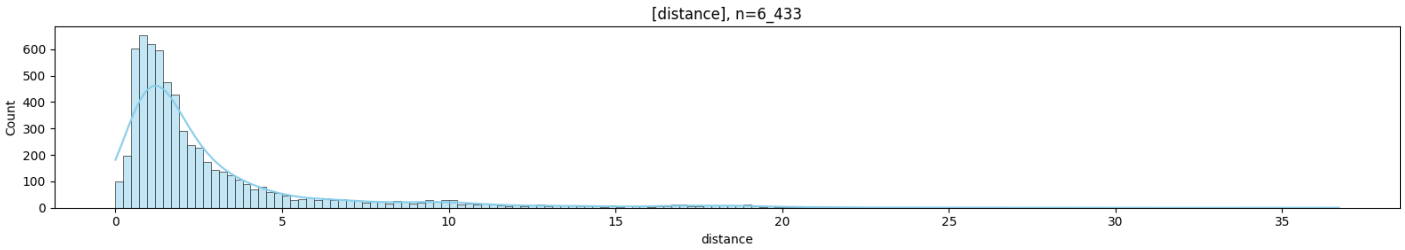
4	2020	18.	13-MV	1
...	...	...	...	...
16352	2023	S37	14-SN	1
16353	2023	S72	11-BE	1
16354	2023	Z22	14-SN	1
16355	2023	Z85	14-SN	1
16356	2023	Z96	14-SN	1

16357 rows × 4 columns

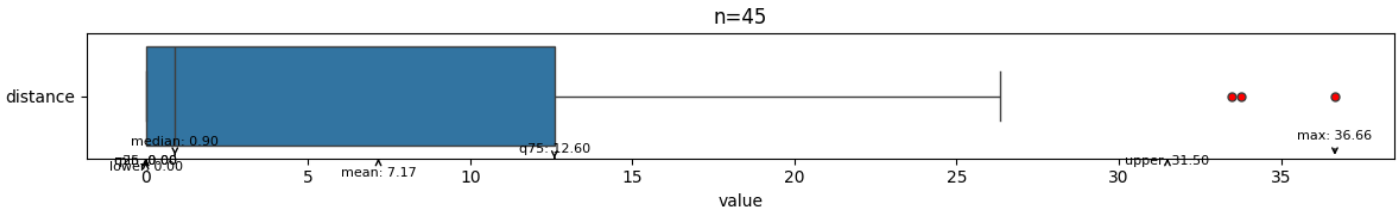
```
z_dy            int64
tu             object
z_kkr_label    object
cnt             int64
dtype: object
```

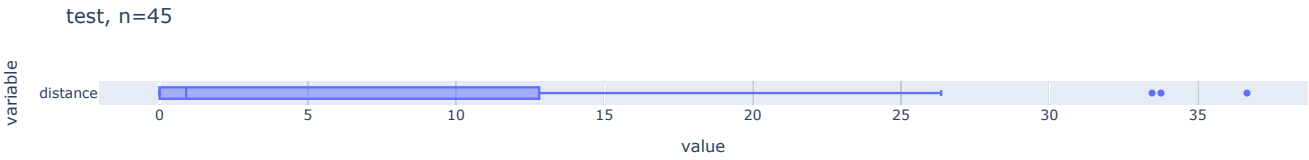
test, n=600_156
01-SH
292 228 158
1_189 1_103 871
6_073 5_380 4_075
02-HH
237 196 200
1_028 849 720
2_984 2_512 1_904
03-NI
7 3 2
34 31 7
160 104 48
04-HB
100 87 30
437 318 201
1_356 1_032 590
05-NW
2_178 1_534 163
9_463 6_816 744
30_681 20_773 2_226
06-HE
25 27 30
796 500 272
927 547 284
847 201 228
814 836
07-RP
75 64 32
381 340 165
555 539 237
312 311 183
1_660 1_442 545
08-BW
1_237 879 470
3_791 3_055 1_558
16_183 12_045 5_726
09-BY
1_774 1_553 1_037
6_167 5_388 3_108
34_546 30_889 19_951
10-SL
165 127 95
640 596 416
2_726 2_395 1_392
11-BE
521 463 402 3
2_434 2_265 1_577
20_698 19_536 16_135
12-BB
499 436 307
16_659 16_623 11_363
13-MV
256 239 169
11_440 9_945 6_603
14-SN
805 648 310 12
31_138 21_507 6_561 280
15-ST
388 272 100
14_655 9_812 4_013
16-TH
135 99 71
5_575 3_416 275 2_092
value
index
col
(other)
C34
C25
C79
C78
C18

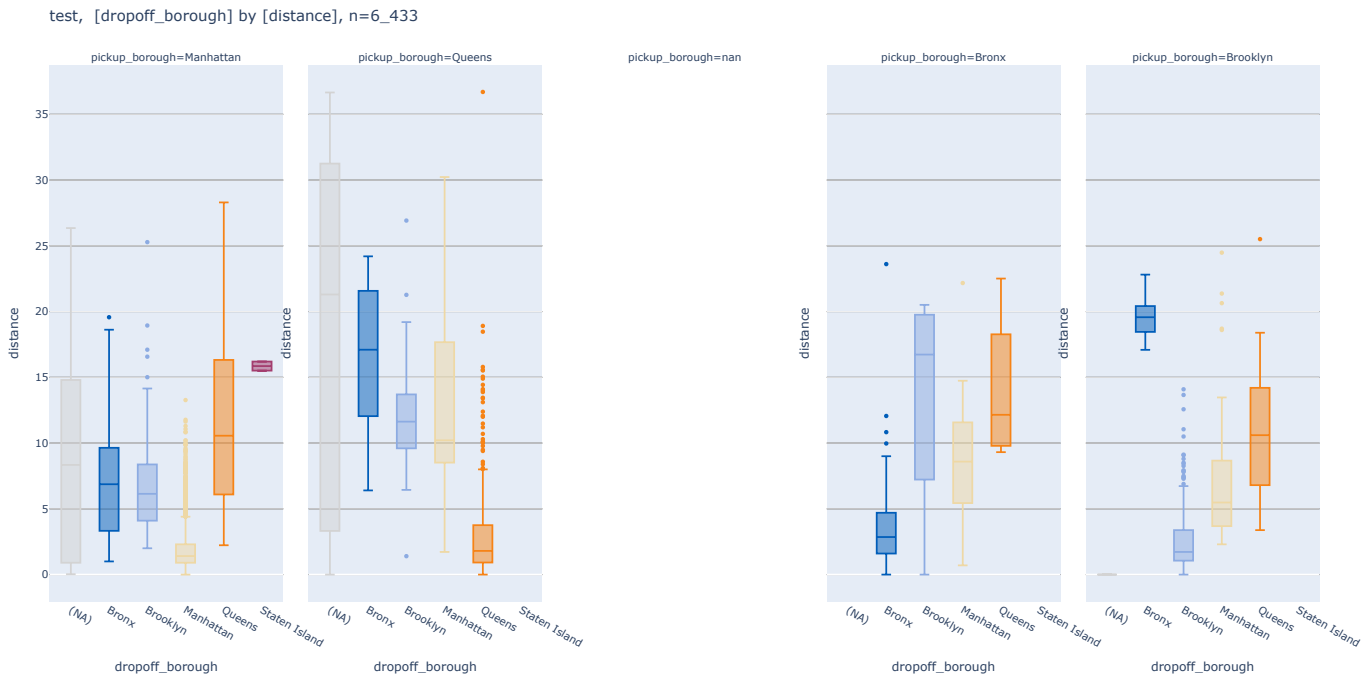
histo


[distance], n=6_433
Count
distance

box

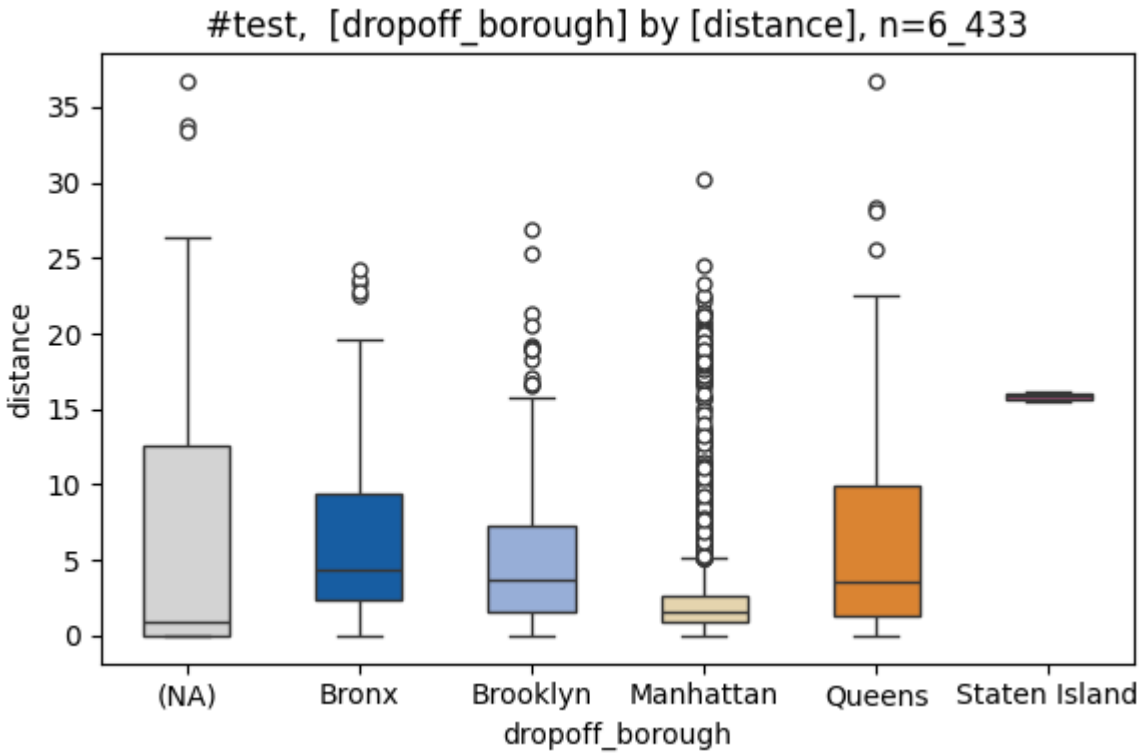




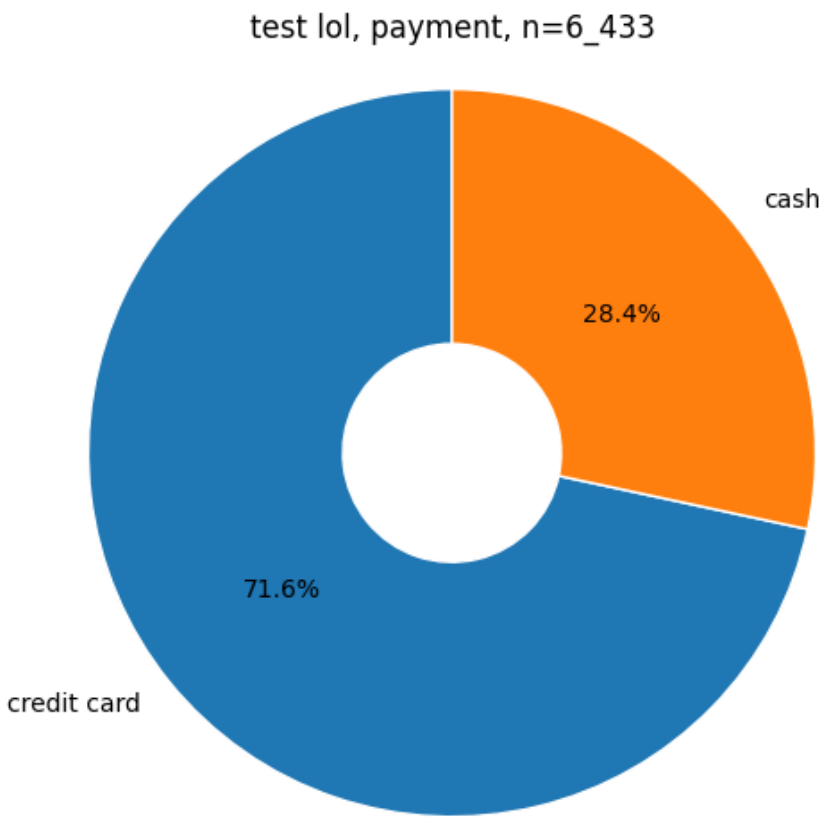



```
     dropoff_borough  distance
0          Manhattan      1.60
1          Manhattan      0.79
2          Manhattan      1.37
3          Manhattan      7.70
4          Manhattan      2.16
...              ...       ...
6428       Manhattan      0.75
6429           Bronx     18.74
6430        Brooklyn      4.14
6431        Brooklyn      1.12
6432        Brooklyn      3.85

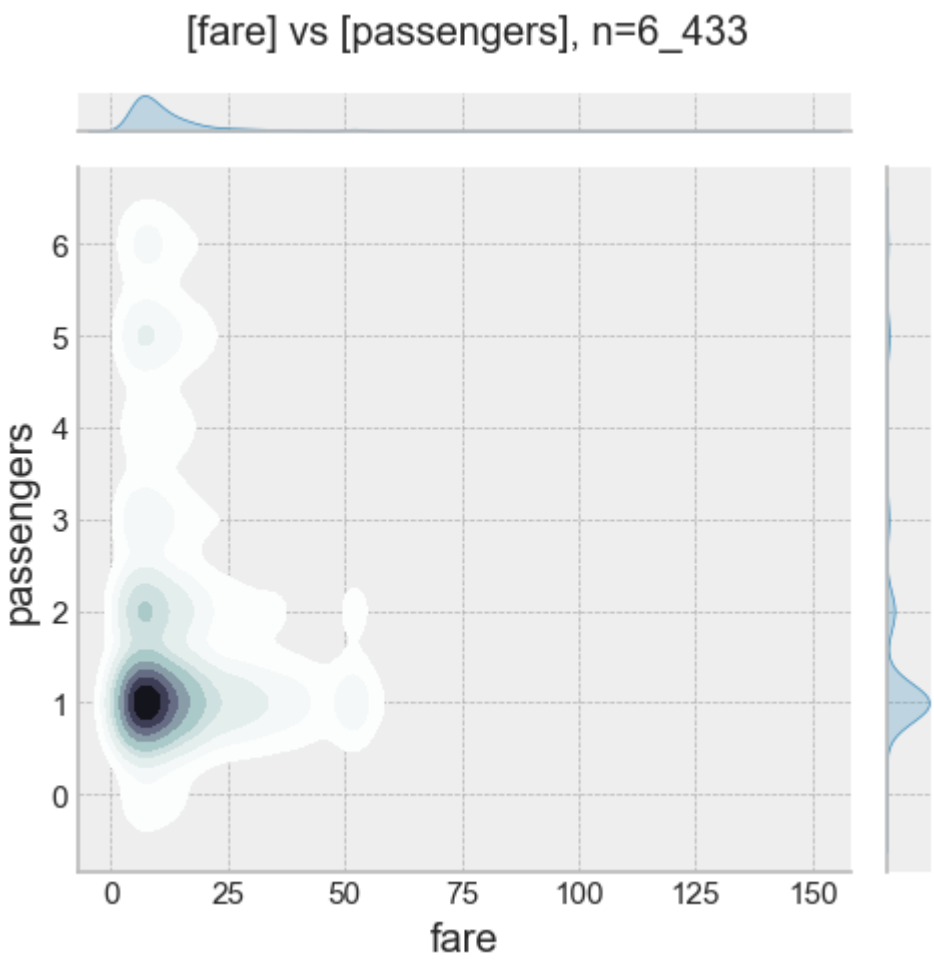
[6433 rows x 2 columns]
```


#test, [dropoff_borough] by [distance], n=6_433
distance
35
30
25
20
15
10
5
0
(NA)
Bronx
Brooklyn
Manhattan
Queens
Staten Island
dropoff_borough

pie


test lol, payment, n=6_433
cash
28.4%
71.6%
credit card

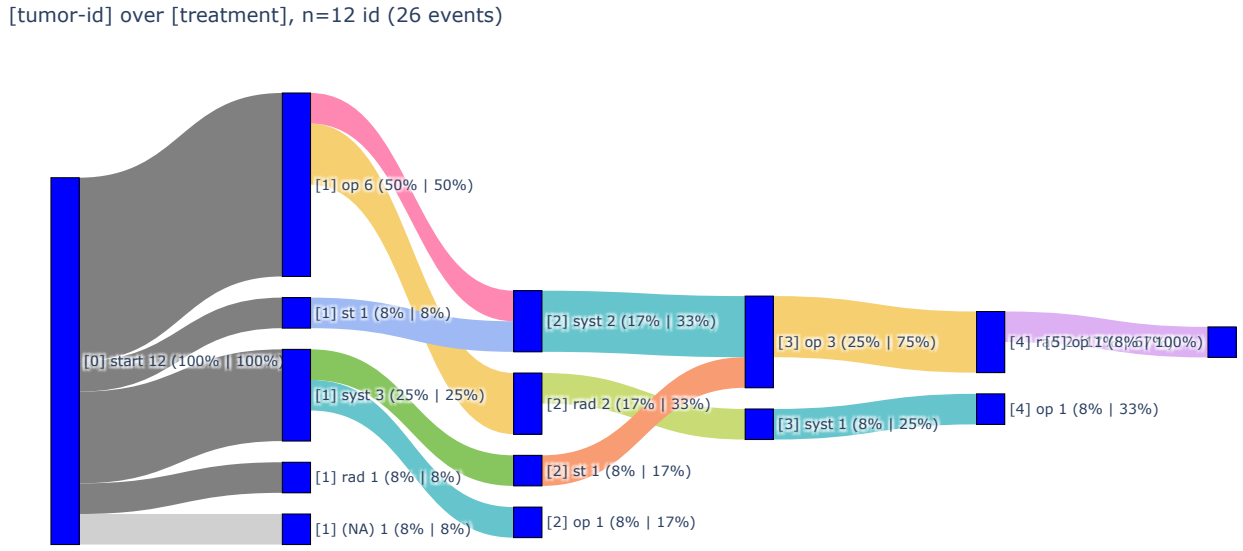
join



sankey

```
--- Using demo data (data_demo) ---
   tumor-id diagnosis date treatment
0         1     2020-01-01        op
1         1     2021-02-01      syst
2         1     2022-03-01        op
3         1     2023-04-01       rad
4         1     2024-05-01        op
5         2     2010-01-01      syst
6         2     2011-02-01        st
7         2     2012-03-01        op
8         2     2013-04-01       rad
9         3     2015-01-01        op
10        3     2016-02-01       rad
11        3     2017-03-01      syst
12        3     2018-04-01        op
13        4     2005-01-01        st
14        4     2006-02-01      syst
15        4     2007-03-01        op
16        5     2019-01-01        op
17        5     2020-02-01       rad
18        6     2021-01-01      syst
19        6     2022-02-01        op
```

```
20         7
21         7
22         8      2025-01-01        op
23         9      2025-02-01        op
24        10      2025-03-01      syst
25        11      2025-04-01       rad
26        12      2025-05-01        op
------------------------------------
```



tbl

print_summary()

```
column (n = 6_433) |   notnull    |  min   | lower |  q25   | median |  mean
|  q75   | upper  |   max   |  std   |  cv   |     sum     | skew  | kurto
-------------------+--------------+-------+-------+--------+--------+--------
+--------+--------+---------+--------+-------+-------------+-------+-------
passengers         | 6_433 (100%) |     0 |     0 |  1.000 |  1.000 |  1.539
|  2.000 |      3 |       6 |  1.204 | 0.782 |       9_902 | 2.359 |  4.881
distance           | 6_433 (100%) | 0.000 | 0.000 |  0.980 |  1.640 |  3.025
|  3.210 |  6.550 |  36.700 |  3.828 | 1.266 |  19_457.360 | 3.008 | 11.135
fare               | 6_433 (100%) | 1.000 | 1.000 |  6.500 |  9.500 | 13.091
| 15.000 | 27.540 | 150.000 | 11.552 | 0.882 |  84_214.870 | 3.217 | 17.649
tip                | 6_433 (100%) | 0.000 | 0.000 |  0.000 |  1.700 |  1.979
|  2.800 |  6.960 |  33.200 |  2.449 | 1.237 |  12_732.320 | 2.665 | 12.740
tolls              | 6_433 (100%) | 0.000 | 0.000 |  0.000 |  0.000 |  0.325
|  0.000 |  0.000 |  24.020 |  1.415 | 4.351 |   2_092.480 | 5.074 | 35.442
total              | 6_433 (100%) | 1.300 | 1.300 | 10.800 | 14.160 | 18.518
| 20.300 | 34.550 | 174.820 | 13.816 | 0.746 | 119_124.970 | 3.093 | 15.053
```

describe_df()

```
🔵 *** df: taxis ***
🟣 shape: (6_433, 14)
🟣 duplicates for ['color']: 6_431 (100%)
🟠 column stats all (dtype | uniques | missings) [values]
- index [0, 1, 2, 3, 4, 5, 6, 7, 8, 9,]
- pickup (datetime64[ns] | 6_414 | 0 (0%)) [2019-02-28 23:29:03, 2019-03-01
00:03:29, 2019-03-01 00:08:32, 2019-03-01 00:15:53,
2019-03-01 00:29:22, 2019-03-01 00:30:59, 2019-03-01 00:32:49, 2019-03-01
00:53:00,
2019-03-01 00:56:50, 2019-03-01 01:25:30,]
- dropoff (datetime64[ns] | 6_425 | 0 (0%)) [2019-02-28 23:32:35, 2019-03-01
00:13:32, 2019-03-01 00:29:47, 2019-03-01 00:32:48,
2019-03-01 00:37:39, 2019-03-01 00:47:58, 2019-03-01 00:53:18, 2019-03-01
00:58:22,
2019-03-01 01:04:01, 2019-03-01 01:29:25,]
- passengers (int64 | 7 | 0 (0%)) [0, 1, 2, 3, 4, 5, 6,]
- distance (float64 | 1_079 | 0 (0%)) [0.0, 0.01, 0.02, 0.03, 0.04, 0.07,
0.08, 0.09, 0.1, 0.11,]
- fare (float64 | 220 | 0 (0%)) [1.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0,
6.5,]
- tip (float64 | 489 | 0 (0%)) [0.0, 0.01, 0.02, 0.06, 0.08, 0.09, 0.1, 0.2,
0.25, 0.33,]
- tolls (float64 | 16 | 0 (0%)) [0.0, 2.64, 4.75, 5.54, 5.75, 5.76, 6.12,
8.4, 8.5, 9.5,]
- total (float64 | 898 | 0 (0%)) [1.3, 3.3, 3.31, 3.8, 4.3, 4.8, 4.81, 5.0,
5.28, 5.3,]
- color (object | 2 | 0 (0%)) ['green', 'yellow',]
- payment (object | 3 | 44 (1%)) ['<NA>', 'cash', 'credit card',]
- pickup_zone (object | 195 | 26 (0%)) ['<NA>', 'Allerton/Pelham Gardens',
'Alphabet City', 'Astoria', 'Auburndale',
'Battery Park', 'Battery Park City', 'Bay Ridge', 'Bayside', 'Bedford',]
- dropoff_zone (object | 204 | 45 (1%)) ['<NA>', 'Allerton/Pelham Gardens',
'Alphabet City', 'Astoria', 'Auburndale',
'Baisley Park', 'Bath Beach', 'Battery Park', 'Battery Park City', 'Bay
Ridge',]
- pickup_borough (object | 5 | 26 (0%)) ['<NA>', 'Bronx', 'Brooklyn',
'Manhattan', 'Queens',]
- dropoff_borough (object | 6 | 45 (1%)) ['<NA>', 'Bronx', 'Brooklyn',
'Manhattan', 'Queens', 'Staten Island',]

🟠 column stats numeric

column (n = 6_433) |   notnull    |  min  | lower |  q25   | median |  mean
|  q75   | upper  |   max   |  std   |  cv
-------------------+--------------+-------+-------+--------+--------+--------
+--------+--------+---------+--------+------
```

```
passengers          | 6_433 (100%) |     0 |     0 |  1.000 |  1.000 |  1.539
|  2.000 |      3 |       6 |  1.204 | 0.782
distance            | 6_433 (100%) | 0.000 | 0.000 |  0.980 |  1.640 |  3.025
|  3.210 |  6.550 |  36.700 |  3.828 | 1.266
fare                | 6_433 (100%) | 1.000 | 1.000 |  6.500 |  9.500 | 13.091
| 15.000 | 27.540 | 150.000 | 11.552 | 0.882
tip                 | 6_433 (100%) | 0.000 | 0.000 |  0.000 |  1.700 |  1.979
|  2.800 |  6.960 |  33.200 |  2.449 | 1.237
tolls               | 6_433 (100%) | 0.000 | 0.000 |  0.000 |  0.000 |  0.325
|  0.000 |  0.000 |  24.020 |  1.415 | 4.351
total               | 6_433 (100%) | 1.300 | 1.300 | 10.800 | 14.160 | 18.518
| 20.300 | 34.550 | 174.820 | 13.816 | 0.746

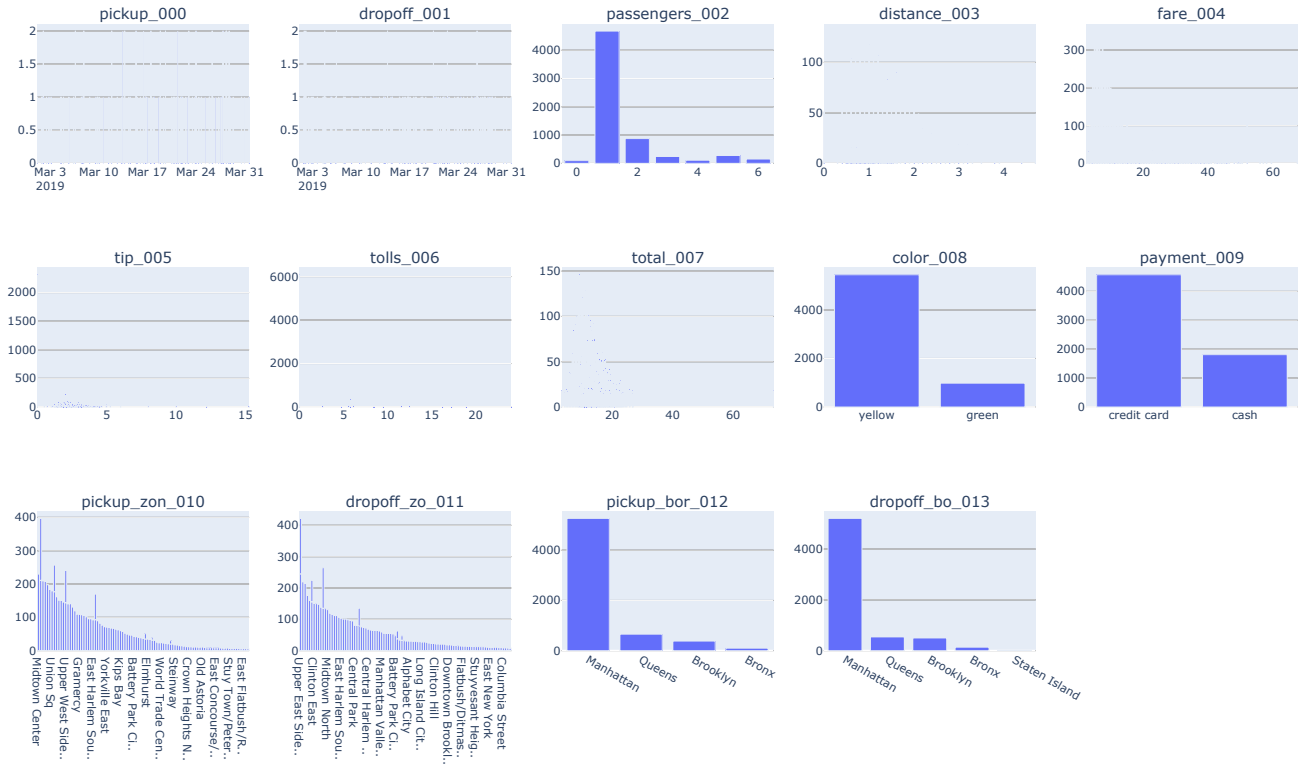

🟠 sample 3 rows
```

```
┌─────────────────────┬─────────────────────┬──────────────┬───┬─────────────────
────────┬─────────────────┬────────────────┐
    │       pickup        │       dropoff       │ passengers │ … │
dropoff_zone     │ pickup_borough │ dropoff_borough │

├─────────────────────┼─────────────────────┼──────────────┼───┼─────────────────
────────┼─────────────────┼────────────────┤
    │ 2019-03-23 20:21:09 │ 2019-03-23 20:27:24 │          1 │ … │ UN/Turtle
Bay South   │ Manhattan       │ Manhattan       │
    │ 2019-03-04 16:11:55 │ 2019-03-04 16:19:00 │          1 │ … │ Upper West
Side So…   │ Manhattan       │ Manhattan       │
    │ 2019-03-27 17:53:01 │ 2019-03-27 18:00:25 │          1 │ … │ West
Village         │ Manhattan       │ Manhattan       │

├─────────────────────┼─────────────────────┼──────────────┼───┼─────────────────
────────┼─────────────────┼────────────────┤
    │ 3 rows
14 columns (6 shown) │

└─────────────────────┴─────────────────────┴──────────────┴───┴─────────────────
────────┴─────────────────┴────────────────┘
```




```
🔵 *** df: <unknown> ***
🟣 shape: (1_000, 105)
🟣 duplicates: 0 (0%)
🟠 column stats all (dtype | uniques | missings) [values]
- index [0, 1, 2, 3, 4,]
- oBDS_RKIPatientTumorId (object | 1_000 | 0 (0%)) ['00427dbf-4acf-468e-9b5c-
8f5f5a7cf1d3', '00e61412-981d-4b2f-b330-6a32e8632f08',
'00eb0411-b445-43cd-8991-4b799ec7873a', '00ee3e10-2024-4ded-9c7a-
d0e606d19767',
'0101ebfe-14d4-4974-9edc-0c04a1c8de43',]
- oBDS_RKIPatientId (object | 886 | 0 (0%)) ['0058eed9-bc0b-44d7-9c82-
ddafed8a39ea', '00ba939d-6368-493d-a652-87e58e7055fd',
'00d3b142-9bea-4a5c-9cad-2d2bf10b7039', '00dffb20-2962-4f9e-ada5-
6bc200e58473',
'00e4c2f0-9c3c-431a-9f13-28010120c376',]
- Diagnosedatum (datetime64[us] | 115 | 0 (0%)) [1900-04-01 00:00:00, 1997-
11-15 00:00:00, 1998-05-15 00:00:00, 1998-12-15 00:00:00,
2000-12-15 00:00:00,]
- Diagnosedatum_Genauigkeit (object | 3 | 0 (0%)) ['M', 'T', 'V',]
- Inzidenzort (object | 287 | 0 (0%)) ['01001', '01002', '01003', '01051',
'01053',]
- Diagnose_ICD10_Code (object | 201 | 0 (0%)) ['C01', 'C02.1', 'C03.0',
'C03.1', 'C04.0',]
- Diagnose_ICD10_Version (object | 13 | 20 (2%)) ['102013GM', '102014GM',
'102015GM', '102016GM', '102017GM',]
- Topographie_Code (object | 181 | 0 (0%)) ['C01.9', 'C02.1', 'C03.0',
'C03.1', 'C04.0',]
```

```
- Topographie_Version (object | 4 | 26 (3%)) ['31', '32', '33', '<NA>',]
- Diagnosesicherung (object | 7 | 0 (0%)) ['0', '1', '2', '5', '6',]
- TNM_Auflage_c (object | 4 | 493 (49%)) ['6', '7', '8', '<NA>',]
- y_Symbol_c (object | 1 | 1_000 (100%)) ['<NA>',]
- r_Symbol_c (object | 1 | 1_000 (100%)) ['<NA>',]
- a_Symbol_c (object | 1 | 1_000 (100%)) ['<NA>',]
- m_Symbol_c (object | 3 | 987 (99%)) ['7', '<NA>', 'm',]
- c_p_u_Praefix_T_c (object | 4 | 749 (75%)) ['<NA>', 'c', 'p', 'u',]
- T_c (object | 24 | 717 (72%)) ['1', '1a', '1b', '1b1', '1c',]
- c_p_u_Praefix_N_c (object | 3 | 730 (73%)) ['<NA>', 'c', 'p',]
- N_c (object | 12 | 700 (70%)) ['0', '1', '1a', '1b', '1c',]
- c_p_u_Praefix_M_c (object | 3 | 714 (71%)) ['<NA>', 'c', 'p',]
- M_c (object | 10 | 683 (68%)) ['0', '1', '1A', '1B', '1C',]
- L_c (object | 4 | 884 (88%)) ['<NA>', 'L0', 'L1', 'LX',]
- V_c (object | 4 | 914 (91%)) ['<NA>', 'V0', 'V1', 'VX',]
- Pn_c (object | 4 | 897 (90%)) ['<NA>', 'Pn0', 'Pn1', 'PnX',]
- S_c (object | 2 | 988 (99%)) ['<NA>', 'SX',]
- UICC_Stadium_c (object | 19 | 863 (86%)) ['0', '0a', '<NA>', 'I', 'IA',]
- TNM_Auflage_p (object | 4 | 333 (33%)) ['6', '7', '8', '<NA>',]
- y_Symbol_p (object | 2 | 989 (99%)) ['<NA>', 'y',]
- r_Symbol_p (object | 1 | 1_000 (100%)) ['<NA>',]
- a_Symbol_p (object | 1 | 1_000 (100%)) ['<NA>',]
- m_Symbol_p (object | 4 | 986 (99%)) ['2', '3', '<NA>', 'm',]
- c_p_u_Praefix_T_p (object | 2 | 515 (52%)) ['<NA>', 'p',]
- T_p (object | 29 | 514 (51%)) ['0', '1', '1(2)', '1C', '1C3',]
- c_p_u_Praefix_N_p (object | 3 | 700 (70%)) ['<NA>', 'c', 'p',]
- N_p (object | 20 | 635 (64%)) ['0', '0(0/18)', '0(0/2)', '0(0/2sn)',
'0(sn)',]
- c_p_u_Praefix_M_p (object | 3 | 771 (77%)) ['<NA>', 'c', 'p',]
- M_p (object | 7 | 694 (69%)) ['0', '1', '1a', '1b', '1c',]
- L_p (object | 4 | 814 (81%)) ['<NA>', 'L0', 'L1', 'LX',]
- V_p (object | 5 | 819 (82%)) ['<NA>', 'V0', 'V1', 'V2', 'VX',]
- Pn_p (object | 4 | 862 (86%)) ['<NA>', 'Pn0', 'Pn1', 'PnX',]
- S_p (object | 3 | 988 (99%)) ['<NA>', 'S1', 'SX',]
- UICC_Stadium_p (object | 21 | 784 (78%)) ['0', '0a', '<NA>', 'I', 'IA',]
- Grading (object | 11 | 0 (0%)) ['0', '1', '2', '3', '4',]
- LK_befallen (Int32 | 13 | 829 (83%)) [0, 1, 2, 3, 4,]
- LK_untersucht (Int32 | 44 | 804 (80%)) [0, 1, 2, 3, 4,]
- Morphologie_Code (object | 153 | 0 (0%)) ['8000/1', '8000/3', '8010/2',
'8010/3', '8010/6',]
- Morphologie_Version (object | 4 | 25 (2%)) ['31', '32', '33', '<NA>',]
- Praetherapeutischer_Menopausenstatus (object | 1 | 1_000 (100%)) ['<NA>',]
- HormonrezeptorStatus_Oestrogen (object | 1 | 1_000 (100%)) ['<NA>',]
- HormonrezeptorStatus_Progesteron (object | 1 | 1_000 (100%)) ['<NA>',]
- Her2neuStatus (object | 1 | 1_000 (100%)) ['<NA>',]
- TumorgroesseInvasiv (Int32 | 1 | 1_000 (100%)) [<NA>,]
- TumorgroesseDCIS (Int32 | 1 | 1_000 (100%)) [<NA>,]
- RASMutation (object | 1 | 1_000 (100%)) ['<NA>',]
- RektumAbstandAnokutanlinie (Int32 | 1 | 1_000 (100%)) [<NA>,]
- GradPrimaer (object | 1 | 1_000 (100%)) ['<NA>',]
- GradSekundaer (object | 1 | 1_000 (100%)) ['<NA>',]
```

```
- ScoreErgebnis (object | 1 | 1_000 (100%)) ['<NA>',]
- AnlassGleasonScore (object | 1 | 1_000 (100%)) ['<NA>',]
- PSA (float32 | 1 | 1_000 (100%)) [nan,]
- DatumPSA (datetime64[us] | 1 | 1_000 (100%)) [NaT,]
- DatumPSA_Genauigkeit (object | 1 | 1_000 (100%)) ['<NA>',]
- Tumordicke (float32 | 1 | 1_000 (100%)) [nan,]
- LDH (Int32 | 1 | 1_000 (100%)) [<NA>,]
- Ulzeration (object | 1 | 1_000 (100%)) ['<NA>',]
- Seitenlokalisation (object | 6 | 0 (0%)) ['B', 'L', 'M', 'R', 'T',]
- DCN (object | 2 | 0 (0%)) ['J', 'N',]
- Anzahl_Tage_Diagnose_Tod (Int32 | 225 | 707 (71%)) [0, 2, 4, 6, 7,]
- z_tum_id (object | 1_000 | 0 (0%)) ['00427dbf-4acf-468e-9b5c-8f5f5a7cf1d3',
'00e61412-981d-4b2f-b330-6a32e8632f08',
'00eb0411-b445-43cd-8991-4b799ec7873a', '00ee3e10-2024-4ded-9c7a-
d0e606d19767',
'0101ebfe-14d4-4974-9edc-0c04a1c8de43',]
- z_kkr (int8 | 16 | 0 (0%)) [1, 2, 3, 4, 5,]
- z_pat_id (object | 886 | 0 (0%)) ['0058eed9-bc0b-44d7-9c82-ddafed8a39ea',
'00ba939d-6368-493d-a652-87e58e7055fd',
'00d3b142-9bea-4a5c-9cad-2d2bf10b7039', '00dffb20-2962-4f9e-ada5-
6bc200e58473',
'00e4c2f0-9c3c-431a-9f13-28010120c376',]
- z_kkr_label (object | 16 | 0 (0%)) ['01-SH', '02-HH', '03-NI', '04-HB',
'05-NW',]
- z_dy (int16 | 27 | 0 (0%)) [1900, 1997, 1998, 2000, 2001,]
- z_age (float64 | 490 | 0 (0%)) [-55.08, -39.67, 2.75, 19.33, 19.67,]
- z_ag05 (object | 17 | 2 (0%)) ['<NA>', 'a00b04', 'a15b19', 'a20b24',
'a25b29',]
- z_icd10 (object | 194 | 0 (0%)) ['C01', 'C02.1', 'C03.0', 'C03.1',
'C04.0',]
- z_icd10_3d (object | 76 | 0 (0%)) ['C01', 'C02', 'C03', 'C04', 'C05',]
- z_t_c_0 (object | 21 | 717 (72%)) ['1', '1a', '1b', '1b1', '1c',]
- z_t_c_1 (object | 8 | 717 (72%)) ['1', '2', '3', '4', '<NA>',]
- z_t_p_0 (object | 23 | 514 (51%)) ['0', '1', '1a', '1a1', '1b',]
- z_t_p_1 (object | 9 | 514 (51%)) ['0', '1', '2', '3', '4',]
- z_n_c_0 (object | 12 | 700 (70%)) ['0', '1', '1a', '1b', '1c',]
- z_n_c_1 (object | 6 | 700 (70%)) ['0', '1', '2', '3', '<NA>',]
- z_n_p_0 (object | 16 | 635 (64%)) ['0', '0(sn)', '1', '1a', '1a(sn)',]
- z_n_p_1 (object | 6 | 635 (64%)) ['0', '1', '2', '3', '<NA>',]
- z_m_c_0 (object | 6 | 686 (69%)) ['0', '1', '1a', '1b', '1c',]
- z_m_c_1 (object | 3 | 686 (69%)) ['0', '1', '<NA>',]
- z_m_p_0 (object | 6 | 701 (70%)) ['0', '1', '1a', '1b', '1c',]
- z_m_p_1 (object | 3 | 701 (70%)) ['0', '1', '<NA>',]
- z_m_pc_1 (object | 3 | 468 (47%)) ['0', '1', '<NA>',]
- z_is_dco (bool | 2 | 0 (0%)) [False, True,]
- z_last_tum_status (object | 10 | 755 (76%)) ['<NA>', 'B - klinische
Besserung des Zustandes', 'D - divergentes Geschehen',
'K - keine Änderung', 'P - Progression',]
- z_tum_op_count (int16 | 6 | 0 (0%)) [0, 1, 2, 3, 4,]
- z_tum_st_count (int16 | 3 | 0 (0%)) [0, 1, 2,]
- z_tum_sy_count (int16 | 6 | 0 (0%)) [0, 1, 2, 3, 4,]
```

```
- z_tum_fo_count (int16 | 14 | 0 (0%)) [0, 1, 2, 3, 4,]
- z_first_treatment (object | 4 | 502 (50%)) ['<NA>', 'op', 'st', 'sy',]
- z_first_treatment_after_days (Int32 | 121 | 502 (50%)) [0, 1, 2, 3, 4,]
- z_event_order (object | 76 | 454 (45%)) ['<NA>', 'fo', 'fo-op', 'fo-op-fo',
'fo-op-fo-sy',]
- z_events (object | 16 | 0 (0%)) ['-', 'fo', 'op', 'op|fo', 'op|st',]
- z_class_hpv (object | 4 | 982 (98%)) ['<NA>', 'N', 'P', 'U',]
- z_tum_order (int8 | 31 | 0 (0%)) [1, 2, 3, 4, 5,]
- z_sex (object | 2 | 0 (0%)) ['M', 'W',]
- z_period_diag_death_day (Int32 | 229 | 700 (70%)) [0, 2, 4, 6, 7,]
- z_period_diag_psa_day (Int32 | 1 | 1_000 (100%)) [<NA>,]

🟠 column stats numeric

column (n = 1_000)           |   notnull    |   min    | lower  |    q25    |
median   |   mean    |    q75    | upper  |   max   |    std    |  cv
-----------------------------+--------------+---------+--------+-----------+-
----------+-----------+-----------+--------+---------+-----------+------
LK_befallen                  |    171 (17%) |       0 |      0 |     0.000 |
0.000 |     1.064 |     1.000 |      2 |      27 |     3.068 | 2.883
LK_untersucht                |    196 (19%) |       0 |      0 |     0.000 |
12.000 |    13.337 |    20.250 |     45 |      80 |    13.506 | 1.013
TumorgroesseInvasiv          |       0 (0%) |     N/A |    N/A |       N/A |
N/A |       N/A |       N/A |    N/A |     N/A |       N/A |   N/A
TumorgroesseDCIS             |       0 (0%) |     N/A |    N/A |       N/A |
N/A |       N/A |       N/A |    N/A |     N/A |       N/A |   N/A
RektumAbstandAnokutanlinie   |       0 (0%) |     N/A |    N/A |       N/A |
N/A |       N/A |       N/A |    N/A |     N/A |       N/A |   N/A
PSA                          |       0 (0%) |     N/A |    N/A |       N/A |
N/A |       N/A |       N/A |    N/A |     N/A |       N/A |   N/A
Tumordicke                   |       0 (0%) |     N/A |    N/A |       N/A |
N/A |       N/A |       N/A |    N/A |     N/A |       N/A |   N/A
LDH                          |       0 (0%) |     N/A |    N/A |       N/A |
N/A |       N/A |       N/A |    N/A |     N/A |       N/A |   N/A
Anzahl_Tage_Diagnose_Tod     |    293 (29%) |       0 |      0 |    55.000 |
261.000 |   701.014 |   720.000 |  1_552 |   8_399 | 1_254.369 | 1.789
z_kkr                        | 1_000 (100%) |       1 |      1 |     5.000 |
7.000 |     7.926 |    12.000 |     16 |      16 |     4.335 | 0.547
z_dy                         | 1_000 (100%) |    1900 |   2017 | 2_020.000 |
2_021.000 | 2_020.463 | 2_022.000 |   2024 |    2024 |     6.198 | 0.003
z_age                        | 1_000 (100%) | -55.080 | 32.420 |    60.310 |
70.670 |    68.287 |    79.250 | 97.920 | 123.580 |    15.605 | 0.229
z_tum_op_count               | 1_000 (100%) |       0 |      0 |     0.000 |
0.000 |     0.469 |     1.000 |      2 |       5 |     0.715 | 1.525
z_tum_st_count               | 1_000 (100%) |       0 |      0 |     0.000 |
0.000 |     0.102 |     0.000 |      0 |       2 |     0.334 | 3.277
z_tum_sy_count               | 1_000 (100%) |       0 |      0 |     0.000 |
0.000 |     0.272 |     0.000 |      0 |       5 |     0.647 | 2.378
z_tum_fo_count               | 1_000 (100%) |       0 |      0 |     0.000 |
0.000 |     0.616 |     0.000 |      0 |      16 |     1.652 | 2.682
z_first_treatment_after_days |    498 (49%) |       0 |      0 |     0.000 |
```

```
12.500 |    39.552 |    36.750 |     91 |   1_091 |    95.339 | 2.410
z_tum_order                  | 1_000 (100%) |       1 |      1 |     1.000 |
1.000 |     1.850 |     1.000 |      1 |      45 |     4.293 | 2.320
z_period_diag_death_day      |    300 (30%) |       0 |      0 |    52.750 |
260.000 |   685.807 |   687.000 |  1_552 |   8_399 | 1_242.187 | 1.811
z_period_diag_psa_day        |       0 (0%) |     N/A |    N/A |       N/A |
N/A |       N/A |       N/A |    N/A |     N/A |       N/A |   N/A


🟠 sample 3 rows
```

```
┌──────────────────────┬──────────────────────┬───┬──────────┬──────────────────
──────┬──────────────────────┐
    │ oBDS_RKIPatientTum…  │  oBDS_RKIPatientId   │ … │  z_sex   │
z_period_diag_deat…  │ z_period_diag_psa_…  │

├──────────────────────┼──────────────────────┼───┼──────────┼──────────────────
──────┼──────────────────────┤
    │ 6f105957-7bbb-41f3…  │ 6fe288a8-24d6-451d…  │ … │ W        │
NULL │                 NULL │
    │ 2334f59e-74ee-477c…  │ c2490ae9-4489-4b9a…  │ … │ W        │
143 │                 NULL │
    │ eb2ae3f0-31a2-4d2f…  │ 32d7f226-18c2-41b8…  │ … │ W        │
0 │                 NULL │

├──────────────────────┴──────────────────────┴───┴──────────┴──────────────────
──────┴──────────────────────┤
    │ 3 rows
105 columns (5 shown) │

└──────────────────────────────────────────────────────────────────────────────
──────────────────────────────┘
```

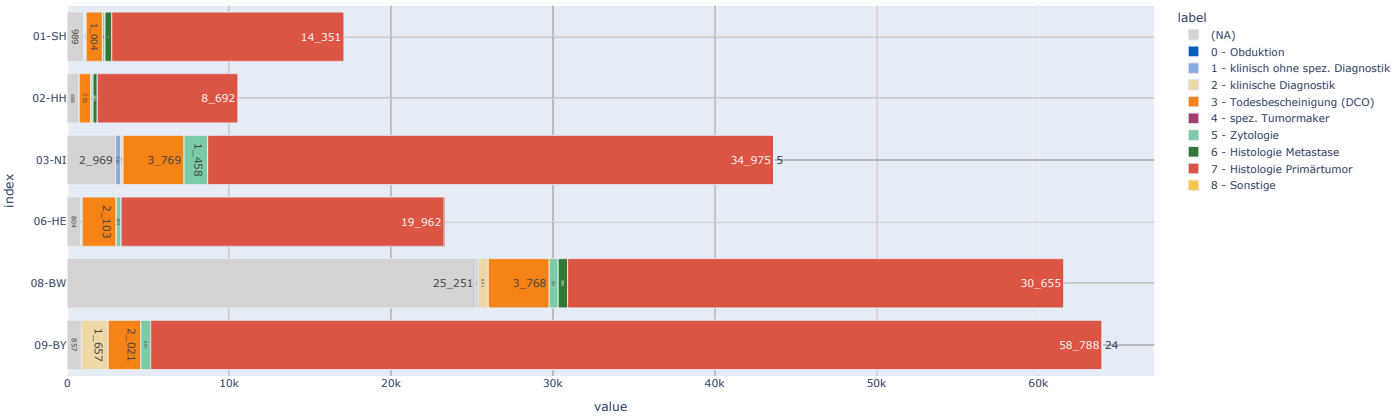
descr_db()

```
    🗄️ delay    1_724, 3
        ("z_kkr_label, z_first_treatment_after_days, z_first_treatment")
```

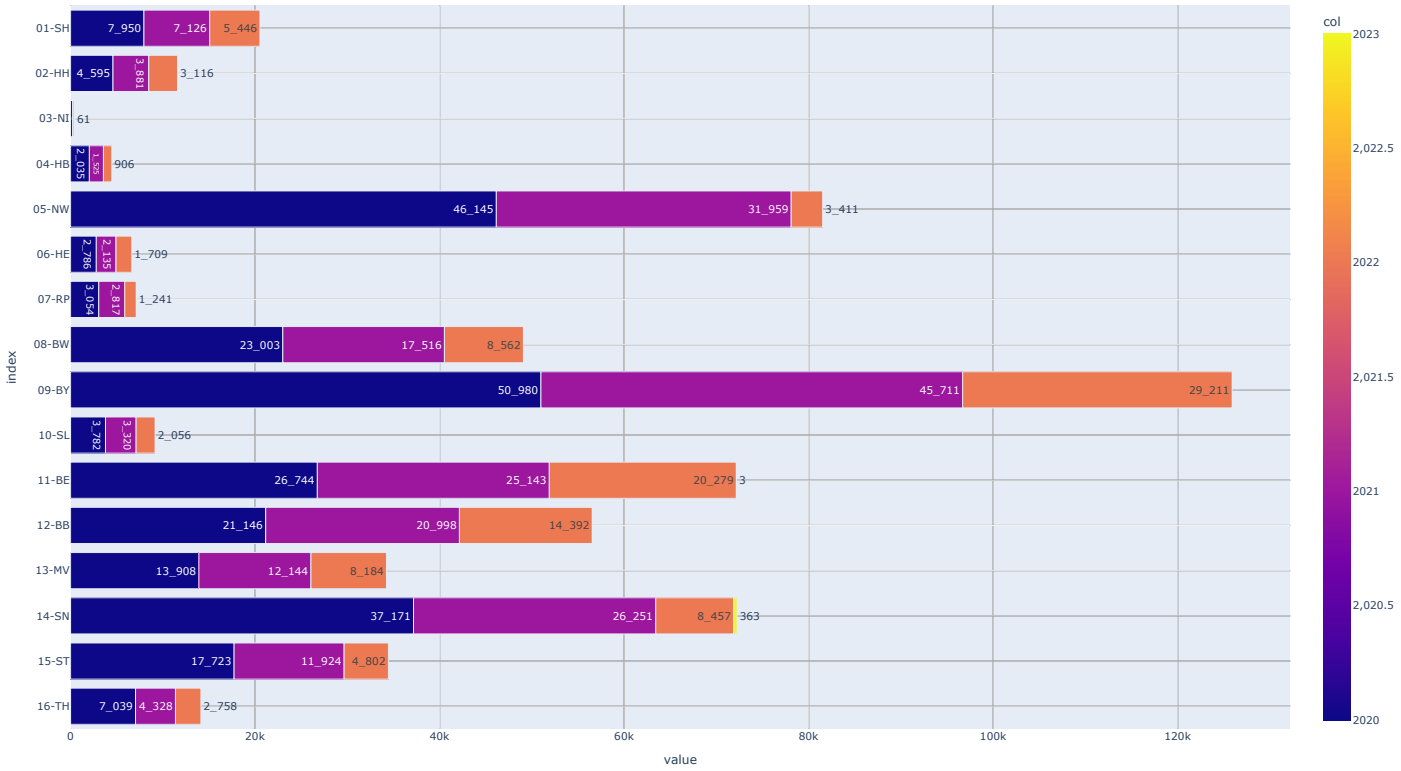
z_kkr_label	z_first_treatment_after_days	z_first_treatment
16-TH	35	sy
05-NW	0	op
13-MV	241	sy

	id	bl	label	cnt
0	0	01-SH	7 - Histologie Primärtumor	14351
1	22	01-SH	3 - Todesbescheinigung (DCO)	1004
2	23	01-SH	5 - Zytologie	146

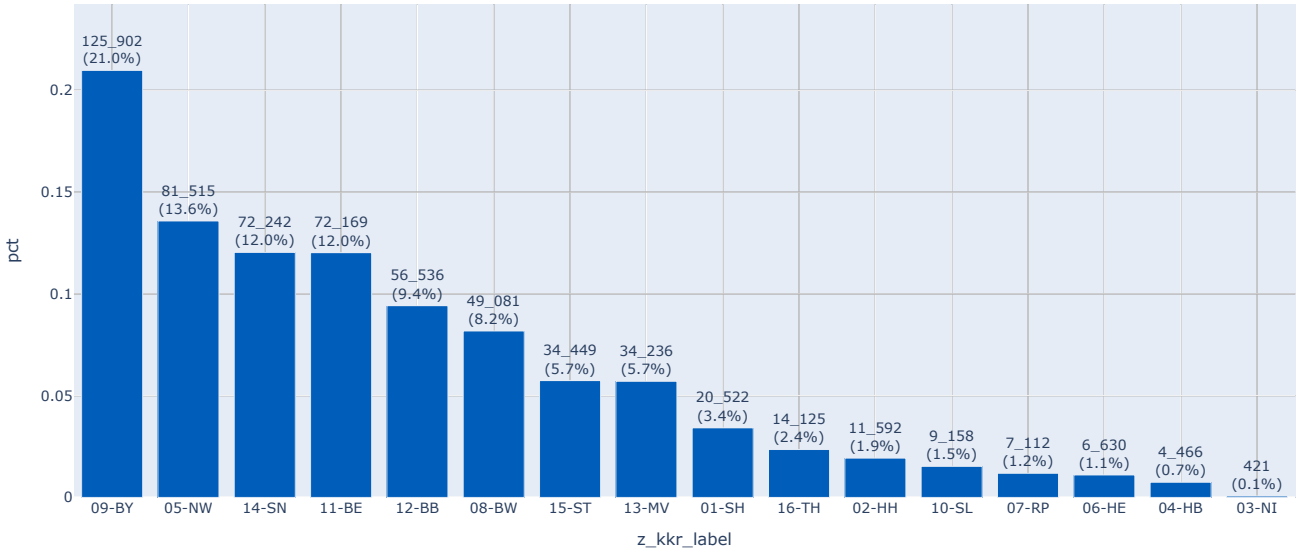
[bl] by [label], n=220_039



[z_kkr_label] by TOP10 [z_dy], n=600_156



[cnt] by [z_kkr_label], n=600_156




```
🔵 *** df: taxis ***
🟣 shape: (6_433, 4)
🟣 duplicates: 4_803 (75%)
🟠 column stats all (dtype | uniques | missings) [values]
- index [0, 1, 2, 3, 4,]
- color (object | 2 | 0 (0%)) ['green', 'yellow',]
- payment (object | 3 | 44 (1%)) ['<NA>', 'cash', 'credit card',]
- pickup_borough (object | 5 | 26 (0%)) ['<NA>', 'Bronx', 'Brooklyn',
'Manhattan', 'Queens',]
```

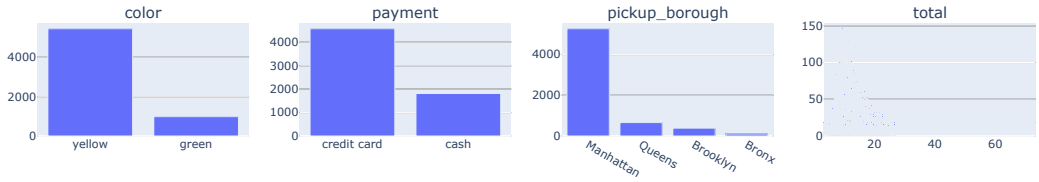
```
- total (float64 | 898 | 0 (0%)) [1.3, 3.3, 3.31, 3.8, 4.3,]

🟠 column stats numeric

column (n = 6_433) |   notnull    |  min  | lower |  q25   | median |  mean
|  q75   | upper  |   max   |  std   |  cv
-------------------+--------------+-------+-------+--------+--------+--------
+--------+--------+---------+--------+------
total              | 6_433 (100%) | 1.300 | 1.300 | 10.800 | 14.160 | 18.518
| 20.300 | 34.550 | 174.820 | 13.816 | 0.746


🟠 sample 3 rows
```

color	payment	pickup_borough	total
yellow	credit card	Manhattan	12.95
yellow	cash	Manhattan	9.3
yellow	credit card	Manhattan	14.16




```
🗄 taxis    6_433, 14
    ("pickup, dropoff, passengers, distance, fare, tip, tolls, total,
color, payment, pickup_zone, dropoff_zone, pickup_borough, dropoff_borough")
```

```
┌────────────────────┬────────────────────┬──────────────┬───┬────────────────
──────┬────────────────────┬────────────────────┐
│       pickup       │       dropoff      │ passengers │ … │
dropoff_zone     │ pickup_borough │ dropoff_borough │

├────────────────────┼────────────────────┼──────────────┼───┼────────────────
```

```
────────┼────────────────────┼────────────────────┼
    │ 2019-03-23 20:21:09 │ 2019-03-23 20:27:24 │          1 │ … │ UN/Turtle
Bay South   │ Manhattan       │ Manhattan       │
    │ 2019-03-04 16:11:55 │ 2019-03-04 16:19:00 │          1 │ … │ Upper West
Side So…   │ Manhattan       │ Manhattan       │
    │ 2019-03-27 17:53:01 │ 2019-03-27 18:00:25 │          1 │ … │ West
Village          │ Manhattan       │ Manhattan       │

├────────────────────┼────────────────────┼──────────────┼───┼──────────────────
────────┼────────────────┼────────────────┼
    │ 3 rows
14 columns (6 shown) │

└────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────┘
```

show num

payment color	(NA)	cash	credit card	Total
green	20 (0.0%)	3_995 (4.7%)	9_774 (11.6%)	13_788 (16.4%)
yellow	508 (0.6%)	17_012 (20.2%)	52_907 (62.8%)	70_427 (83.6%)
Total	528 (0.6%)	21_006 (24.9%)	62_681 (74.4%)	84_215 (100.0%)

	distance	fare	tip	tolls	Total
0	1.600 (2.0%)	7.000 (8.7%)	2.150 (2.7%)	0	10.750 (13.3%)
1	0.790 (1.0%)	5.000 (6.2%)	0	0	5.790 (7.2%)
2	1.370 (1.7%)	7.500 (9.3%)	2.360 (2.9%)	0	11.230 (13.9%)
3	7.700 (9.5%)	27.000 (33.4%)	6.150 (7.6%)	0	40.850 (50.5%)
4	2.160 (2.7%)	9.000 (11.1%)	1.100 (1.4%)	0	12.260 (15.2%)
Total	13.620 (16.8%)	55.500 (68.6%)	11.760 (14.5%)	0	80.880 (100.0%)

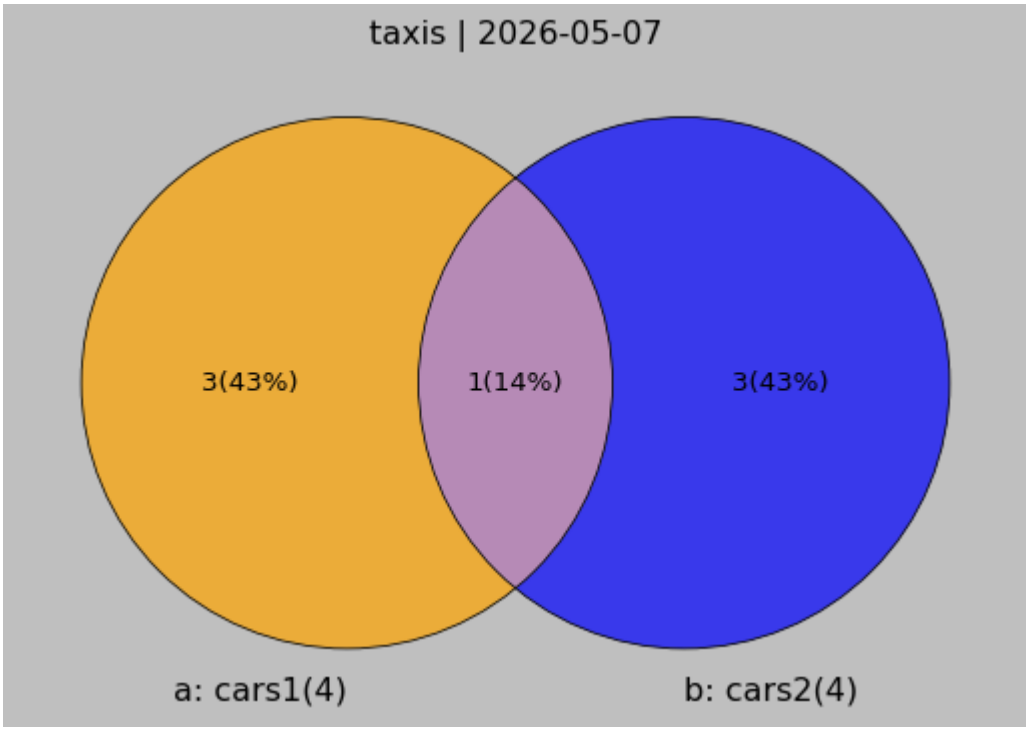
	z_kkr_label	z_first_treatment
0	05-NW	None
1	05-NW	None
2	09-BY	sy
3	09-BY	sy
4	05-NW	None
...	...	...
656	09-BY	sy
657	02-HH	op
658	05-NW	None
659	02-HH	sy
660	09-BY	sy

661 rows × 2 columns

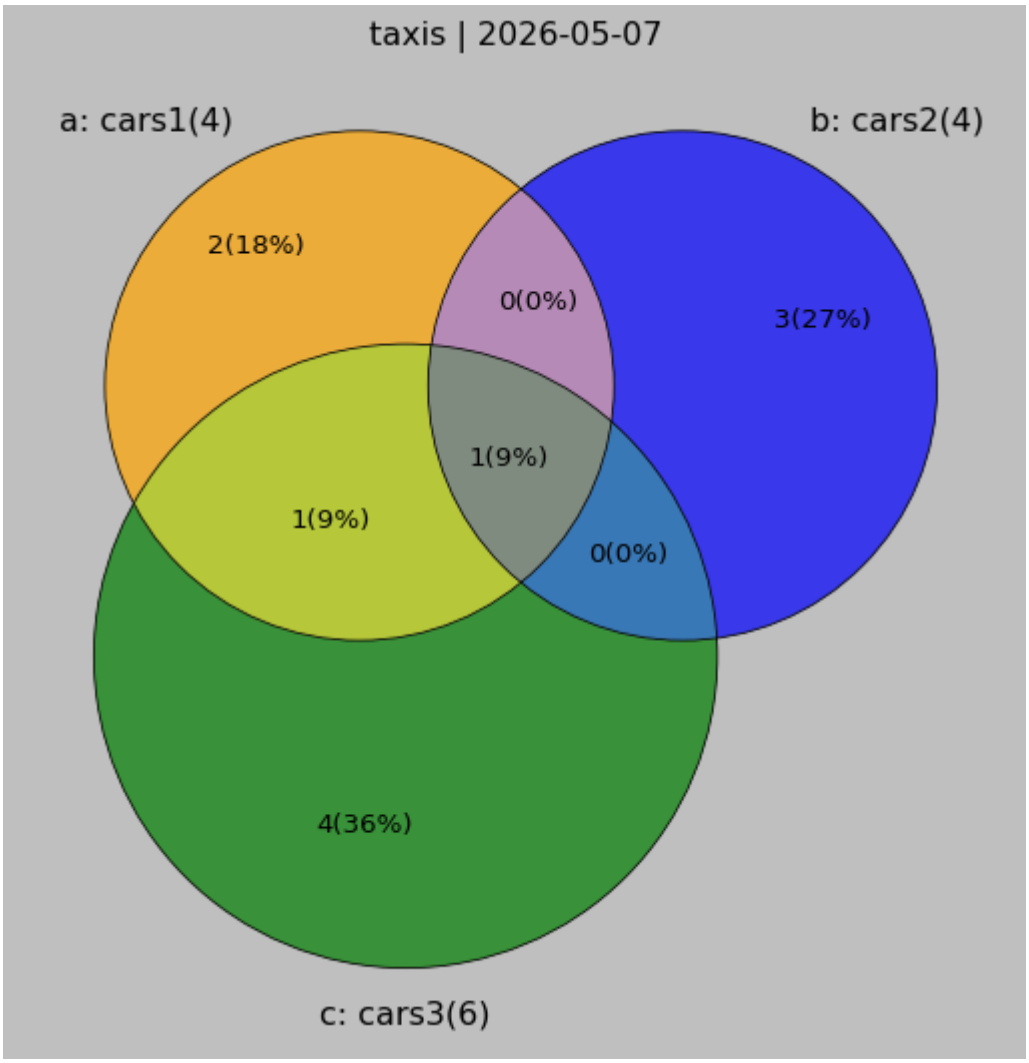
z_first_treatment z_kkr_label	(NA)	op	st	sy	Total
01-SH	0	0	0	0	0
02-HH	0	18.000 (2.7%)	4.000 (0.6%)	28.000 (4.2%)	50.000 (7.6%)
03-NI	0	0	0	0	0
04-HB	0	0	0	0	0
05-NW	352.000 (53.3%)	0	0	0	352.000 (53.3%)
06-HE	0	0	0	0	0
07-RP	0	0	0	0	0
08-BW	0	0	0	0	0
09-BY	0	11.000 (1.7%)	28.000 (4.2%)	186.000 (28.1%)	225.000 (34.0%)
10-SL	0	0	0	0	0
11-BE	0	0	0	0	0
12-BB	0	0	0	0	0
13-MV	0	0	0	0	0
14-SN	0	0	0	0	0
15-ST	0	2.000 (0.3%)	1.000 (0.2%)	31.000 (4.7%)	34.000 (5.1%)
16-TH	0	0	0	0	0
Total	352.000 (53.3%)	31.000 (4.7%)	33.000 (5.0%)	245.000 (37.1%)	661.000 (100.0%)

venn

```
ab --> cars1 | cars2 --> len: 7
```



```
abc --> cars1 | cars2 | cars3 --> len: 11
ab --> cars1 | cars2 --> len: 7
ac --> cars1 | cars3 --> len: 8
bc --> cars2 | cars3 --> len: 9
```



hlp

```
    -- filter-sql
        z_dy between 2020 and 2024
        and z_kkr_label is not null
        and z_icd10_3d in ('C18','C19','C20')
        and
        (
            left(Code,7) in ('5-455.2', '5-455.4', '5-455.7')
            OR left(Code,5) in ('5-987')
        )
```

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"\n    \n    z_dy between 2020 and 2024\n    and z_kkr_label is not null\n\n
and z_icd10_3d in ('C18','C19','C20')\n    and --sql\n    (\n
left(Code,7) in ('5-455.2', '5-455.4', '5-455.7')\n        OR left(Code,5) in
('5-987')\n    )\n\n"
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## 📊 Sample Data & Explanation
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### Original Sample DataFrame
This data represents various measurements (like 'Distance') grouped by a
category ('Borough').
The goal is to restructure the data so that the 'Distance' values for each
'Borough' are in separate, comparable columns.
  Borough  Distance  ID
Manhattan       1.5 101
   Queens       5.1 102
Manhattan       0.8 103
    Bronx       3.2 104
   Queens       2.5 105
Manhattan       1.4 106
    Bronx       7.8 107
   Queens       4.0 108
```

```
### Sparse Pivoted DataFrame
The data has been **pivoted**:
* The non-numeric column **'Borough'** is used to create the new column names
('Bronx', 'Manhattan', 'Queens').
* The numeric column **'Distance'** provides the values for the new columns.
* A new index (**'Distance_sequence'**) is created to align the measurements
within each 'Borough' group, allowing comparison of the 1st, 2nd, 3rd, etc.,
measurement across boroughs.
* `NaN` values appear where a borough has fewer measurements than others.
Borough            Bronx  Manhattan  Queens
Distance_sequence
0                    3.2        1.5     5.1
1                    7.8        0.8     2.5
2                    NaN        1.4     4.0
```

```
**Summary Statistics:**
           count      mean       std  min   25%  50%   75%  max
Borough
Bronx        2.0  5.500000  3.252691  3.2  4.35  5.5  6.65  7.8
Manhattan    3.0  1.233333  0.378594  0.8  1.10  1.4  1.45  1.5
Queens       3.0  3.866667  1.305118  2.5  3.25  4.0  4.55  5.1
```

Borough Distance_sequence	Bronx	Manhattan	Queens
0	3.2	1.5	5.1
1	7.8	0.8	2.5
2	NaN	1.4	4.0

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|  165429 |
    | 5-401.11 - Exzision einzelner Lymphknoten und Lymphgefäße: Axillär: Mit
Radionuklidmarkierung (Senti │  135826 |
    | 5-573.40 - Transurethrale Inzision, Exzision, Destruktion und Resektion
von (erkranktem) Gewebe der  │   94266 |
    | 5-987.0 - Anwendung eines OP-Roboters: Komplexer OP-Roboter
|   82761 |
    | 5-870.a1 - Partielle (brusterhaltende) Exzision der Mamma und
Destruktion von Mammagewebe: Partielle │   65040 |
    | 5-870.a2 - Partielle (brusterhaltende) Exzision der Mamma und
Destruktion von Mammagewebe: Partielle │   52410 |
    | 5-984 - Mikrochirurgische Technik
|   47640 |
    | 5-604.52 - Radikale Prostatovesikulektomie: Laparoskopisch, gefäß- und
nervenerhaltend: Mit regional │   39369 |
    | 5-895.14 - Radikale und ausgedehnte Exzision von erkranktem Gewebe an
Haut und Unterhaut: Ohne primä │   35085 |
    | 5-573.41 - Transurethrale Inzision, Exzision, Destruktion und Resektion
von (erkranktem) Gewebe der  │   34869 |

├──────────────────────────────────────────────────────────────────────────────
──────────────────────────────┴──────────┤
    | 10 rows
2 columns |

└──────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────┘
```

```
0      2019-03-23 20:21:09
1      2019-03-04 16:11:55
2      2019-03-27 17:53:01
3      2019-03-10 01:23:59
4      2019-03-30 13:27:42
               ...
6428   2019-03-31 09:51:53
6429   2019-03-31 17:38:00
6430   2019-03-23 22:55:18
6431   2019-03-04 10:09:25
6432   2019-03-13 19:31:22
Name: pickup, Length: 6433, dtype: datetime64[ns]
```

```
    tum_id: d90eb5e3-2791-420e-9682-69a8e9dfff9a
    pat
```

```
┌──────────────────────┬─────────┬────────┬───┬──────────────────────┬──────────
──────────┬──────────────────────┐
│       z_pat_id       │  z_sex  │ z_age  │ … │ Geburtsdatum_Genau…  │
Datum_Vitalstatus │ Datum_Vitalstatus_…  │
├──────────────────────┼─────────┼────────┼───┼──────────────────────┼──────────
──────────┼──────────────────────┤
│ cb00a5de-22fd-495b…  │ M       │  69.92 │ … │ T                    │ 2024-
10-15        │ T                    │
├──────────────────────┼─────────┼────────┼───┼──────────────────────┼──────────
──────────┼──────────────────────┤
│ 1 rows
9 columns (6 shown) │
└──────────────────────────────────────────────────────────────────────────────
──────────────────────────────────┘
```

```
    tod
```

```
┌──────────────────┬──────────┬──────────┬──────────────────┐
│ TodesursacheId │   Code   │ Version │ IsGrundleiden │
├──────────────────┼──────────┼──────────┼──────────────────┤
│                       0 rows                              │
└──────────────────────────────────────────────────────────┘
```

```
tum1
```

```
┌──────────────┬──────────┬─────────────────┬───┬─────────────────┬──────────────────────────┬──────────────────────────┬──────────
```

```
┌──────────┬────────────────────┬────────────────────┬────────────────┬──────────────────
──────┬──────────────────────┬──────────┐
│ Grading │ Morphologie_Code │ Topographie_Code │ Inzidenzort │
Diagnosesicherung │ Seitenlokalisation │   DCN   │
├──────────┼────────────────────┼────────────────────┼────────────────┼──────────────────
──────┼──────────────────────┼──────────┤
│ 3       │ 8140/3           │ C61.9            │ 13071       │ 7
│ T                  │ N       │
└──────────┴────────────────────┴────────────────────┴────────────────┴──────────────────
──────┴──────────────────────┴──────────┘
```

```
    tum4
```

```
┌──────────┬──────────┬──────────┬───┬────────────────────────┬────────────────────
──┬────────────────────────┐
│ z_t_p_0 │ z_n_p_0 │ z_m_p_0 │ … │ Praetherapeutische…  │
HormonrezeptorStat…  │ HormonrezeptorStat…  │
├──────────┼──────────┼──────────┼───┼────────────────────────┼────────────────────
──┼────────────────────────┤
│ NULL    │ NULL    │ NULL    │ … │ NULL                   │ NULL
│ NULL                 │
├──────────┼──────────┼──────────┼───┼────────────────────────┼────────────────────
──┼────────────────────────┤
│ 1 rows
8 columns (6 shown) │
└──────────────────────────────────────────────────────────────────────────────────────
──────────────────────────┘
```

```
    tum5
```

```
┌──────────────────────┬────────────────────┬────────────────┬────────────────┬──────
──┬──────────────┬────────┬──────────────┐
│ TumorgroesseInvasiv │ TumorgroesseDCIS │ RASMutation │ ScoreErgebnis │  PSA
│ Tumordicke │  LDH  │ Ulzeration │
├──────────────────────┼────────────────────┼────────────────┼────────────────┼──────
──┼──────────────┼────────┼──────────────┤
```

```
│                 NULL │              NULL │ NULL         │ 8              │
130.0 │       NULL │  NULL │ NULL        │
└──────────────────────┴───────────────────┴──────────────┴────────────────┴──────
──┴────────────┴───────┴──────────────┘
```

```
    op
```

```
┌──────────────────────┬────────────┬──────────────────────┬───┬────────────────
────────┬──────────────┐
│         OPId         │ Intention │ Lokale_Beurteilung…  │ … │
z_period_diag_op_day │ z_op_order │
├──────────────────────┼────────────┼──────────────────────┼───┼────────────────
────────┼──────────────┤
│ ec494ef0-e73d-4762…  │ K         │ RX                   │ … │
897 │          1 │
├──────────────────────┼────────────┼──────────────────────┼───┼────────────────
────────┼──────────────┤
│ 1 rows
8 columns (5 shown) │
└──────────────────────────────────────────────────────────────────────────────
────────────────────┘
```

```
    ops
```

```
┌──────────────────────────────────────┬──────────┬─────────┬──────────────────
──────────────────────┐
│                OPSId                 │   Code   │ Version │
OP_TypId               │
├──────────────────────────────────────┼──────────┼─────────┼──────────────────
──────────────────────┤
│ a1f94049-78be-444d-b13d-10c02a79327a │ 5-031.30 │ 2022    │ ec494ef0-e73d-
4762-b022-7227aaf2ff57 │
└──────────────────────────────────────┴──────────┴─────────┴──────────────────
──────────────────────┘
```

```
    st
```

```
┌──────────────────────────────────────┬──────────────┬──────────────────┐
│                  STId                │  Intention   │  Stellung_OP     │
├──────────────────────────────────────┼──────────────┼──────────────────┤
│ 56b19ab4-3c34-4c31-85e3-2945199a8b2b │ P            │ 0                │
│ 72a7f8c1-4436-440a-a646-e9bbaea91663 │ P            │ 0                │
│ 825ace54-695f-42a6-8b9d-0bd05165268f │ P            │ 0                │
│ 8fd82c06-6a60-4419-983a-c18c9441a544 │ P            │ 0                │
│ 17f45990-667f-49cb-a178-b95a98e5e8ff │ P            │ 0                │
│ 3334fc97-99ef-4657-9780-277b07c7785f │ P            │ 0                │
│ 337d17d0-25fd-4f70-aeeb-3e2a8543d0c4 │ P            │ 0                │
│ e6629ef7-13d3-48fc-a4c3-4fd9f7bb773e │ P            │ 0                │
│ ea82aeb8-6ade-421c-8185-e20cae5d927e │ P            │ 0                │
│ ef7b0226-d577-401a-ae01-808f5613730a │ K            │ 0                │
│ eff1fce1-16a1-4806-b7d0-f41ffec9ed98 │ S            │ S                │
├──────────────────────────────────────┴──────────────┴──────────────────┤
│ 11 rows                                                   3 columns     │
└─────────────────────────────────────────────────────────────────────────┘
```

```
    be
```

```
┌──────────────────────┬──────────────────────┬──────────────────────┬───┬──────
──────────────────────┬──────────────┐
│    BestrahlungId     │ Anzahl_Tage_Diagno…  │ Anzahl_Tage_ST_Dauer │ … │
z_period_diag_best…  │ z_bestr_order │
├──────────────────────┼──────────────────────┼──────────────────────┼───┼──────
──────────────────────┼──────────────┤
│ b121f45a-221e-481d…  │                   43 │                    3 │ … │
43 │             1 │
│ 4cd066be-9355-4067…  │                   49 │                   25 │ … │
49 │             2 │
│ 9e2ea731-0140-4f35…  │                  854 │                    4 │ … │
854 │             3 │
│ 041d9288-6c70-4f53…  │                  996 │                   13 │ … │
996 │             4 │
│ 8ab778d9-7244-42a0…  │                 1172 │                    0 │ … │
1172 │             5 │
│ b185569c-7972-4635…  │                 1226 │                    0 │ … │
1226 │             6 │
│ 3d0960de-30b5-4188…  │                 1281 │                    0 │ … │
1281 │             7 │
│ d9404472-f9e3-47c4…  │                 1333 │                    0 │ … │
```

```
1333 │             8 │
│ 8a6aa060-d578-487c…  │                 1333 │                    0 │ … │
1333 │             9 │
│ 43a1aff7-5898-4da1…  │                 1386 │                    0 │ … │
1386 │            10 │
│ b33226f3-b6cd-4d96…  │                 1386 │                    0 │ … │
1386 │            11 │
│ 570b5f4d-0ec2-4cc9…  │                 1442 │                    0 │ … │
1442 │            12 │
│ e024423c-d480-49b9…  │                 1442 │                    0 │ … │
1442 │            13 │
│ db6c9dbf-06d1-4276…  │                 1556 │                    0 │ … │
1556 │            14 │
│ 995fc348-feee-482a…  │                 1556 │                    0 │ … │
1556 │            15 │
├──────────────────────┼──────────────────────┼──────────────────────┼───┼────
──────────────────┴────────────────┤
│ 15 rows
8 columns (5 shown) │
└──────────────────────────────────────────────────────────────────────────────
──────────────────────────────────┘
```

```
    app
```

```
┌──────────────────────┬──────────────────────┬───┬──────────────────┬──────────
────────┬──────────────────┐
│    BestrahlungId     │ TypeOfST_TypBestra…  │ … │ Atemgetriggert │
CodeVersion2014 │ CodeVersion2021 │
├──────────────────────┼──────────────────────┼───┼──────────────────┼──────────
────────┼──────────────────┤
│ b121f45a-221e-481d…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 313
│ NULL            │
│ 4cd066be-9355-4067…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 541
│ NULL            │
│ 9e2ea731-0140-4f35…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 67
│ NULL            │
│ 041d9288-6c70-4f53…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 65
│ NULL            │
│ 8ab778d9-7244-42a0…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 81
│ NULL            │
│ b185569c-7972-4635…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 81
│ NULL            │
│ 3d0960de-30b5-4188…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 81
│ NULL            │
│ d9404472-f9e3-47c4…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 6
```

```
│ NULL            │
│ 8a6aa060-d578-487c…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 542
│ NULL            │
│ 43a1aff7-5898-4da1…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 542
│ NULL            │
│ b33226f3-b6cd-4d96…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 6
│ NULL            │
│ 570b5f4d-0ec2-4cc9…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 542
│ NULL            │
│ e024423c-d480-49b9…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 6
│ NULL            │
│ db6c9dbf-06d1-4276…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 542
│ NULL            │
│ 995fc348-feee-482a…  │ ST_TypBestrahlungA…  │ … │ NULL           │ 6
│ NULL            │
├────────────────────────┼────────────────────────┼───┼──────────────────┼──────────
────────┼──────────────────┤
│ 15 rows
11 columns (5 shown) │
└──────────────────────────────────────────────────────────────────────────────────
──────────────────────────┘
```

```
    syst
```

```
┌────────────────────────┬────────────┬──────────────┬───┬────────────────────────┬
──────────────────────┬────────────────┐
│        SYSTId          │ Intention │ Stellung_OP │ … │ Datum_Beginn_SYST_…  │
z_period_diag_syst…  │ z_syst_order │
├────────────────────────┼────────────┼──────────────┼───┼────────────────────────┼
──────────────────────┼────────────────┤
│ 73aa6922-bc09-4af1…  │ S          │ 0            │ … │ T                      │
29 │             1 │
│ e38d4eed-ad2d-4f6f…  │ S          │ 0            │ … │ T                      │
57 │             2 │
│ 2624813f-392d-4600…  │ S          │ 0            │ … │ T                      │
141 │             3 │
│ df3ac679-6da8-4678…  │ P          │ 0            │ … │ T                      │
688 │             4 │
│ 077140e2-e752-4bb9…  │ P          │ 0            │ … │ T                      │
862 │             5 │
│ 102bf12e-be20-47de…  │ P          │ 0            │ … │ T                      │
869 │             6 │
│ d2717912-2fa2-4eb8…  │ P          │ 0            │ … │ T                      │
1045 │             7 │
│ 102c8567-ec63-4ac7…  │ P          │ 0            │ … │ T                      │
```

```
1073 │             8 │
├────────────────────┼─────────┼──────────────┼───┼────────────────────┼
────────────────────┼──────────────┤
│ 8 rows
10 columns (6 shown) │
└──────────────────────────────────────────────────────────────────────
──────────────────────────────────────┘
```

```
    subst
```

```
┌────────────────────┬────────────────────┬────────────────────┬───┬───
──────────────────┬────────────────────┐
│      SubstanzId      │ TypeOfSYST_TypSubs…  │ SYST_TypSubstanzAT…  │ … │
z_substance_predic…  │ z_substance_predic…  │
├────────────────────┼────────────────────┼────────────────────┼───┼───
──────────────────┼────────────────────┤
│ 2f5803c7-df95-4f64…  │ Bezeichnung          │ NULL                 │ … │
Docetaxel            │                  1.0 │
│ 7c3b7991-ada9-4f92…  │ Bezeichnung          │ NULL                 │ … │
Cabazitaxel          │                  1.0 │
│ f802f4ba-562e-4053…  │ Bezeichnung          │ NULL                 │ … │
Enzalutamid          │                  1.0 │
│ 478a99ec-4942-47da…  │ Bezeichnung          │ NULL                 │ … │
Zoledronsäure        │                  1.0 │
│ 6e49d638-8b4d-4c7c…  │ Bezeichnung          │ NULL                 │ … │
Denosumab            │                  1.0 │
├────────────────────┼────────────────────┼────────────────────┼───┼───
──────────────────┼────────────────────┤
│ 5 rows
8 columns (5 shown) │
└──────────────────────────────────────────────────────────────────────
──────────────────────────────────────┘
```

```
    prot
```

```
┌────────────────────┬────────────────────┬────────────────────┬──────
──────────────┬──────────────────────────────┐
│        SYSTId        │ TypeOfProtokoll_Typ │     Bezeichnung      │
Protokoll_TypProto…  │ Protokoll_TypProtokolls…  │
```

```
├──────────────────────┼──────────────────────┼──────────────────────┼────────
──────────────┼──────────────────────────┤
│ 73aa6922-bc09-4af1…  │ Bezeichnung          │ Bicalutamidflareup   │ NULL
│ NULL                 │
│ e38d4eed-ad2d-4f6f…  │ Bezeichnung          │ Leuprorelin->fortl…  │ NULL
│ NULL                 │
│ 2624813f-392d-4600…  │ Bezeichnung          │ Abirateron+Prednis…  │ NULL
│ NULL                 │
│ df3ac679-6da8-4678…  │ Bezeichnung          │ Docetaxel            │ NULL
│ NULL                 │
│ 077140e2-e752-4bb9…  │ Bezeichnung          │ Cabazitaxelfortl.    │ NULL
│ NULL                 │
│ 102bf12e-be20-47de…  │ Bezeichnung          │ Enzalutamidfortl.    │ NULL
│ NULL                 │
│ d2717912-2fa2-4eb8…  │ Bezeichnung          │ Zoledronsäurefortl.  │ NULL
│ NULL                 │
│ 102c8567-ec63-4ac7…  │ Bezeichnung          │ Denosumabfortl.      │ NULL
│ NULL                 │
└──────────────────────┴──────────────────────┴──────────────────────┴────────
──────────────┴──────────────────────────┘
```

```
    fo
```

```
┌──────────────────────┬──────────────────────┬───┬──────────────────────┬────
──────────┬──────────────────────┐
│   FolgeereignisId    │ Gesamtbeurteilung_…  │ … │ Datum_Folgeereigni…  │
z_fo_order │ z_period_diag_fo_day │
├──────────────────────┼──────────────────────┼───┼──────────────────────┼────
──────────┼──────────────────────┤
│ 00f3ce83-391f-4c4a…  │ K                    │ … │ T                    │
1 │                  123 │
│ 98f85314-5e08-4447…  │ K                    │ … │ T                    │
2 │                  214 │
│ 28b5a28c-32b9-4b2d…  │ K                    │ … │ T                    │
3 │                  549 │
│ 38ce200a-9dbe-4d52…  │ P                    │ … │ T                    │
4 │                  641 │
│ f9784935-338a-4099…  │ P                    │ … │ T                    │
5 │                  883 │
│ d0b9b58a-485e-4053…  │ P                    │ … │ T                    │
6 │                  975 │
│ f2a1bcd8-691f-4ec6…  │ P                    │ … │ T                    │
7 │                 1156 │
├──────────────────────┼──────────────────────┼───┼──────────────────────┼────
──────────┼──────────────────────┤
```

```
│ 7 rows
9 columns (5 shown) │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

```
fo_tnm
```

```
┌──────────────────────────────────┬──────────────────────────────────┬──────────┬──────────┬───┬──────────────────────────────────────────────────────────────────────────────────────────────────────────────┬──────────┬──────────┬──────────────┐
│                TNMId             │          FolgeereignisId         │ Version  │ y_Symbol │ … │                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
│ 00f3ce83-391f-4c4a…  │ 00f3ce83-391f-4c4a…  │ NULL     │ NULL     │ … │ NULL     │ NULL     │ NULL         │
│ 98f85314-5e08-4447…  │ 98f85314-5e08-4447…  │ NULL     │ NULL     │ … │ NULL     │ NULL     │ NULL         │
│ 28b5a28c-32b9-4b2d…  │ 28b5a28c-32b9-4b2d…  │ NULL     │ NULL     │ … │ NULL     │ NULL     │ NULL         │
│ 38ce200a-9dbe-4d52…  │ 38ce200a-9dbe-4d52…  │ NULL     │ NULL     │ … │ NULL     │ NULL     │ NULL         │
│ f9784935-338a-4099…  │ f9784935-338a-4099…  │ NULL     │ NULL     │ … │ NULL     │ NULL     │ NULL         │
│ d0b9b58a-485e-4053…  │ d0b9b58a-485e-4053…  │ NULL     │ NULL     │ … │ NULL     │ NULL     │ NULL         │
│ f2a1bcd8-691f-4ec6…  │ f2a1bcd8-691f-4ec6…  │ NULL     │ NULL     │ … │ NULL     │ NULL     │ NULL         │
├──────────────────────────────────┼──────────────────────────────────┼──────────┼──────────┼───┼──────────┼──────────┼──────────────┤
│ 7 rows
18 columns (8 shown) │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

```
fo_fm
```

```
┌──────────────────┬────────────────────┬──────────────────┐
│ FolgeereignisId  │ FernmetastaseId    │ Lokalisation     │
├──────────────────┼────────────────────┼──────────────────┤
```

```
│                        0 rows                        │
└──────────────────────────────────────────────────────┘
```

```
fo_weitere
```

```
┌──────────────────────────┬──────────┬─────────┬────────────────────┐
│ WeitereKlassifikationId │  Name    │ Stadium │ FolgeereignisId    │
├──────────────────────────┼──────────┼─────────┼────────────────────┤
│                          0 rows                                    │
└────────────────────────────────────────────────────────────────────┘
```

```
diag_fm
```

```
┌──────────────────────────────────────┬────────────────┐
│           FernmetastaseId            │ Lokalisation   │
├──────────────────────────────────────┼────────────────┤
│ 2cc01989-99cd-4ff2-9165-622f6021282d │ OSS            │
└──────────────────────────────────────┴────────────────┘
```

```
diag_weitere
```

```
┌──────────────────────────┬──────────┬─────────┐
│ WeitereKlassifikationId │  Name    │ Stadium │
├──────────────────────────┼──────────┼─────────┤
│                    0 rows                    │
└──────────────────────────────────────────────┘
```

```
['total', 'distance']


⏳ Adding datetime columns basing off of: dropoff
```

	pickup	dropoff	passengers	distance	fare	tip	tolls	total	color	payment	...	dropoff_zone	pickup_borough	dropoff_borough	YYYY	MM	Q	YYYY-MM	YYYYQ	YYYY-WW	DDD
0	2019-03-23 20:21:09	2019-03-23 20:27:24	1	1.60	7.0	2.15	0.0	12.95	yellow	credit card	...	UN/Turtle Bay South	Manhattan	Manhattan	2019	3	1	2019-03	2019Q1	2019-W12	Sat
1	2019-03-04 16:11:55	2019-03-04 16:19:00	1	0.79	5.0	0.00	0.0	9.30	yellow	cash	...	Upper West Side South	Manhattan	Manhattan	2019	3	1	2019-03	2019Q1	2019-W10	Mon
2	2019-03-27 17:53:01	2019-03-27 18:00:25	1	1.37	7.5	2.36	0.0	14.16	yellow	credit card	...	West Village	Manhattan	Manhattan	2019	3	1	2019-03	2019Q1	2019-W13	Wed
3	2019-03-10 01:23:59	2019-03-10 01:49:51	1	7.70	27.0	6.15	0.0	36.95	yellow	credit card	...	Yorkville West	Manhattan	Manhattan	2019	3	1	2019-03	2019Q1	2019-W10	Sun
4	2019-03-30 13:27:42	2019-03-30 13:37:14	3	2.16	9.0	1.10	0.0	13.40	yellow	credit card	...	Yorkville West	Manhattan	Manhattan	2019	3	1	2019-03	2019Q1	2019-W13	Sat
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
6428	2019-03-31 09:51:53	2019-03-31 09:55:27	1	0.75	4.5	1.06	0.0	6.36	green	credit card	...	Central Harlem North	Manhattan	Manhattan	2019	3	1	2019-03	2019Q1	2019-W13	Sun
6429	2019-03-31 17:38:00	2019-03-31 18:34:23	1	18.74	58.0	0.00	0.0	58.80	green	credit card	...	East Concourse/Concourse Village	Queens	Bronx	2019	3	1	2019-03	2019Q1	2019-W13	Sun
6430	2019-03-23 22:55:18	2019-03-23 23:14:25	1	4.14	16.0	0.00	0.0	17.30	green	cash	...	Bushwick North	Brooklyn	Brooklyn	2019	3	1	2019-03	2019Q1	2019-W12	Sat
6431	2019-03-04 10:09:25	2019-03-04 10:14:29	1	1.12	6.0	0.00	0.0	6.80	green	credit card	...	East Flatbush/Remsen Village	Brooklyn	Brooklyn	2019	3	1	2019-03	2019Q1	2019-W10	Mon
6432	2019-03-13 19:31:22	2019-03-13 19:48:02	1	3.85	15.0	3.36	0.0	20.16	green	credit card	...	Windsor Terrace	Brooklyn	Brooklyn	2019	3	1	2019-03	2019Q1	2019-W11	Wed

6433 rows × 21 columns

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[The most important themes of the draft referendum are:
1. Improving the use of health data for research and innovation particularly
through
the establishment of a National Data Access Point;
2. Enhancing the coordination and networking of different stakeholders
involved
in health data management;
3. Expanding the scope of the General Data Protection Regulation (GDPR) to
cover all
areas of healthcare;
4. Providing better access to health data for patients and researchers
including through
the development of a new law on health data protection;
5. Facilitating the exchange of health data between different countries and
regions
particularly within the European Union (EU);
6. Ensuring that data are used in a way that promotes sustainable development
and achieves
the United Nations Sustainable Development Goals (SDGs).]
```